

To Follow the Crowd? Benefits and Costs of Migrant Networks*

Yulu Tang[†]
LSE

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Abstract

Globally, migrant workers often cluster with hometown peers in the same sectors and locations. Using data on millions of domestic migrant workers from a food delivery platform in China, I quantify two countervailing effects of migrant networks: learning benefits and congestion costs. First, I provide direct evidence of peer learning. New workers learn from their peers' past delivery experiences, reducing search time by 10%. Using quasi-random variation in workers' location choices induced by entry bonuses, I show that having one hometown peer nearby increases new workers' productivity by 2%. Second, I quantify a new congestion cost of clustering due to correlated shocks: during adverse hometown shocks, clustered workers simultaneously increase labor supply. With inelastic labor demand, hometown workers compete for deliveries and real wages decrease by 10%. I build an equilibrium location choice model to quantify the trade-off between the learning benefits and congestion costs. Counterfactuals show that providing insurance for hometown shocks doubles clustering levels and increases productivity.

Keywords: Migrant Network, Learning from Peers, Risk Sharing

JEL Classification: O12, O15, R23, J61, D83

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[†]London School of Economics. Email: y.tang82@lse.ac.uk

1 Introduction

Throughout the world, migrant workers often cluster with hometown peers in the same sectors and locations. For example, 14% of taxi drivers in New York City came from Pakistan (Schaller, 2006); 24% of hairdressers in Houston were Vietnamese migrants (Patel and Vella, 2013). This pattern also holds among domestic migrants. For example, nearly 30% of migrant workers from the same county choose to work in the same sector and province in China (Chen et al., 2010).

What are the benefits and costs of migrant clustering? Does joining a large hometown network improve or harm migrant workers' welfare? Answering these questions is crucial for understanding how clustering fundamentally shapes migrant workers' economic outcomes and informing migration policies. However, there are at least two major empirical challenges in the literature. First, it is difficult to observe migrant workers' labor market performance and spatial sorting at high frequency and granular geographic scale (see Munshi (2020) for a comprehensive review). Second, while both benefits and costs matter for worker welfare, few studies can jointly examine them in the same context.

In this paper, I provide a unified theoretical and empirical analysis of two countervailing effects of migrant networks. First, I show how clustering facilitates peer learning, increasing productivity and average wages. Second, I identify a new congestion cost: clustering causes hometown workers to compete with one another during adverse hometown shocks when their labor supply surges simultaneously. This mechanism reduces real wages precisely when workers have the greatest desire to work longer, amplifying income volatility. I use an equilibrium model to analyze how these two forces interact to affect workers' location choices and welfare.

To overcome the empirical challenges outlined above, I use rich microdata from a major food delivery platform in China covering millions of domestic migrant workers.¹ This setting offers two key advantages. First, rich GPS data, which tracks delivery workers' real-time movements, allows for precise measurement of worker productivity and peer learning. Second, I observe clustering at a highly granular level: hometown networks within individual neighborhoods in a destination city. Moreover, with the gig economy growing rapidly, it has become the top occupational choice among migrant workers (Pew, 2021), yet we know relatively little about the choices made by these gig-economy workers. By focusing on this setting, this project provides new results on worker welfare under this new economic form.

The main analysis consists of three parts. In part 1, I provide direct evidence of the learning benefits of clustering. New workers increase productivity by learning from peers' past delivery experiences. In part 2, I identify the labor market congestion cost of clustering due to correlated hometown shocks. In part 3, I build an equilibrium location choice model to quantify the trade-off between the learning benefits and congestion costs and their combined impacts on worker welfare.

¹The platform operates similarly to companies such as *Uber Eats* and *DoorDash*. Delivery workers earn a piece rate by delivering meals from restaurants to consumers.

Before proceeding to the main analysis, I provide three key facts about the magnitude of migrant clustering in the context. First, 98% of delivery workers in large cities, such as Beijing and Shanghai, are domestic migrants, primarily born in lower-GDP provinces in western and central China. Second, delivery workers typically operate within small neighborhoods rather than across entire cities. Most workers concentrate their activity within a single $5\text{km} \times 5\text{km}$ district, where they complete over 70% of their daily deliveries. Third, clustering by hometown is substantial within each district: on average, 30% of workers from the same county of birth choose to work in the same district—a very high clustering level compared to random allocation where fewer than 1% would co-locate. As a result, the largest hometown cluster accounts for 15% of a district’s total delivery workforce on the platform on average.

In Part 1, I conduct two main analyses to quantify the learning benefits of clustering. First, I open the black box of peer learning by identifying the types of knowledge shared among workers. Second, I use quasi-random variation in workers’ location choices induced by entry bonuses to estimate the aggregate impact of clustering on productivity.

In the first analysis, I examine whether delivery performance improves when workers or their peers have prior experience with a given location. Specifically, I ask whether workers deliver faster (1) when they themselves have previously visited the location, capturing learning by doing, and (2) when their peers have previously visited the location, capturing peer learning.² The identification strategy relies on the platform’s order allocation algorithm, which exogenously assigns workers to visit different locations. While the algorithm is not fully random, its primary input is each worker’s real-time location, so that the platform can match orders with nearby workers. I thus control for delivery-level assignment probabilities by reshuffling orders among workers who are at the same place and time.³ This strategy implies a quasi-random assignment: workers at the same location at the same time are close substitutes in the algorithm’s assignment process. Consequently, which worker receives which order is approximately random among eligible workers at that location.

Using high-frequency GPS data, I decompose each delivery into three components: (a) time searching for the restaurant, (b) travel time on main roads, and (c) time searching for the customer’s location. This decomposition allows me to identify exactly where in the delivery process learning occurs. I find that both learning by doing and peer learning reduce search times at restaurants and customer locations. A worker’s own prior visit to a location reduces search time by 20% on subsequent deliveries. Peer knowledge is roughly half as effective. However, travel speed on main roads does not display the same patterns. These results align with field insights that the platform’s GPS tools can guide workers along main roads, but accumulated experience is essential for navigating local neighborhoods and solving other last-mile delivery problems. Additional heterogeneity analysis shows that learning effects are larger in older and denser neighborhoods.

²I identify a worker’s closest peer by using the network data linking each new worker to his referrer, who is also an existing delivery worker. Around 60% of new workers have a referrer when they join the platform the first time.

³I follow the intuition from [Borusyak and Hull \(2023\)](#). I provide more details on implementation in Section 3.2.

In the second analysis, I quantify the overall impact of clustering on worker productivity. A key empirical challenge is that workers' location choices are endogenous, which may bias OLS estimates. To address this, I construct an instrumental variable using quasi-random variation in new workers' location choices induced by entry bonuses. These entry bonuses are fixed monetary rewards offered to new workers who complete a minimum number of deliveries in bonus-target districts.⁴ These bonuses thus act as natural experiments that exogenously incentivize new workers to enter bonus-offering districts.⁵ The endogenous variable is the hometown clustering level (the share of workers from the same county) in a worker's actual work district. The instrument is the clustering level in the bonus-target district within the worker's choice set.⁶ I also restrict to bonuses offered after a worker registers, ensuring bonuses affect district choice conditional on joining the platform. I discuss the full IV construction in Section 3.

The IV estimates show that clustering significantly increases new workers' productivity. On average, each additional hometown peer in the district increases a new worker's productivity by 0.4% over the first three months. Working in a district with an average clustering level increases productivity by 5%. While clustering does not significantly affect hours worked, total earnings increase proportionally with productivity. To explore nonlinearity in the clustering–productivity relationship, I estimate an IV regression with indicators for different clustering levels. The results reveal an increasing but concave pattern: productivity increases with clustering, but at a diminishing rate. This diminishing marginal effect is consistent with knowledge spillovers saturating as networks cover most locations within a district.

In Part 2, I identify a new congestion cost of clustering. The core intuition is that **during adverse hometown shocks, migrant workers tend to increase labor supply to send remittances home** (e.g., [Kochar, 1999](#)). When workers cluster by hometown, many affected workers increase labor supply simultaneously. When labor demand is inelastic in the destination labor market, clustering increases competition and decreases real wages, precisely when workers most need higher earnings. Clustering thus amplifies rather than smooths income volatility.

To test this mechanism, I construct two measures of hometown shocks: (1) county-level floods between June and August 2020—the worst since 1998; and (2) county-level COVID-19 lockdowns in 2021. Both shocks produce similar patterns. First, workers from affected counties increase their weekly labor supply by six hours on average. This allows them to earn around 20 dollars more per week, which is likely to be remitted to support their families back home.

Second, I construct indicators of aggregate shock in destination districts based on the predicted

⁴The platform occasionally offers entry bonuses in specific districts and weeks to balance local labor supply and demand. In Shanghai in 2021, 72% of entry bonuses were active for only one week, and 87% of districts offered at least one bonus week during the year.

⁵While the assignment of bonuses is not random, my identification strategy does not require it to be. I include district $\times$ week fixed effect in the main specification. This means I compare workers who both face the same entry bonus and work in the same district, but differ in the share of hometown peers that they have in the bonus-target district. I elaborate on this point in section 3.3.

⁶I define each worker's choice set as districts within 6 km of their referrer's district. 90% of new workers choose to work in districts within this choice set.

share of workers from shock-affected counties in each district and week between June and August 2020.⁷ I find that districts with high exposure to hometown shocks, defined as having more than 15% of workers from affected counties, experience 10% increases in total labor supply. In contrast, consumer demand, measured by total order volume, remains unchanged.

Finally, I estimate the congestion effect by regressing workers' weekly labor market performances on individual hometown shock indicator, aggregate district shock indicator, and their interaction. The interaction term captures the congestion effect of experiencing a hometown shock while working in a high-clustering district. It shows that despite working longer hours during hometown shocks, clustered workers complete significantly fewer deliveries than non-clustered workers facing the same shock. On average, their real wages fall by 10%. This mechanism is increased competition for a fixed pool of orders. When many clustered workers simultaneously extend their hours, the number of active workers per hour surges, reducing the number of deliveries assigned per worker.⁸

In Part 3, I build an equilibrium model to quantify the trade-off between these two effects of clustering: the first-moment productivity gains from peer learning and the second-moment income risks from labor market congestion.⁹ The model consists of four stages. First, conditional on migrating to the destination city and becoming a delivery worker, each worker chooses their desired work district that maximizes expected utility. Next, adverse hometown shocks are realized with some probability. Third, workers choose working hours and remittances to maximize utility, where they care about their own consumption, their left-behind family's consumption, and leisure. Finally, equilibrium wages are determined to clear the labor market in each district. The model assumes that workers are aware of both the learning and congestion effects of clustering and incorporate them into their location decisions.

I estimate six sets of key model parameters using empirical results from Steps 1 and 2, supplemented by additional quasi-experimental variation and model-implied moment conditions. First, productivity gains from clustering come directly from the IV estimates in Step 1. Second, the magnitude of hometown shocks is identified from workers' labor supply responses to floods in Step 2. Third, I estimate workers' labor supply elasticity using a quasi-experiment. During "worker festivals," the platform randomly awards prizes ranging from 50 to 1,000 RMB. I regress working hours on prize amounts to recover the elasticity of labor supply with respect to income.

There are six sets of key parameters in the model. I estimate them using a combination of empirical results from Steps 1 and 2, additional quasi-experiments, and model-implied moment conditions. First, productivity gains from clustering come directly from the IV estimates in Step 1.

⁷I compute the predicted shares of workers using the hometown-district composition in May 2020 as a baseline. Using predicted rather than actual shares helps avoid bias from endogenous worker sorting in response to shocks.

⁸I also examine whether clustered workers respond by relocating to other districts. I only find weak increases in relocation rates. One likely explanation is that after relocating, workers experience slower delivery speed.

⁹A structural model is essential to credibly evaluate this trade-off, as it requires estimating workers' risk aversion and other utility function parameters. In addition, the model enables tracing the full utility-clustering relationship, quantifying externalities, and conducting policy-relevant counterfactuals.

Second, the magnitude of hometown shocks is identified from workers' labor supply responses to floods in Step 2. Third, I estimate workers' labor supply elasticity using a quasi-experiment. During "worker festivals" organized by the platform, they randomly select workers and awards prizes ranging from 50 to 1,000 RMB. I regress working hours on lottery amounts to recover the elasticity of labor supply with respect to income.

Two additional key parameters are estimated from model-predicted moments. First, I identify the risk aversion coefficient from differences in clustering levels across hometowns with varying shock probabilities. The model predicts that workers from riskier hometowns shall cluster less to avoid greater exposure to congestion costs. Moreover, this gap in clustering levels increases with risk aversion. I therefore use observed differences in clustering between high-risk and low-risk hometowns to recover the risk aversion parameter. Second, I estimate the scale of workers' idiosyncratic location preferences from a gravity regression of district entry choices on entry bonuses. Finally, I calibrate the average probability of hometown shocks using historical rainfall data from 2000 to 2020.

With these estimated parameters, I find an inverted-U relationship between clustering and worker utility: utility initially rises with clustering due to learning benefits but eventually declines as congestion costs dominate. The model also predicts two sources of externalities: (1) a positive learning externality, as workers do not internalize how their own knowledge benefits peers; and (2) a negative congestion externality, as excess clustering causes all workers to wait longer for delivery assignments during hometown shocks. On net, I find the cumulative externality to be negative. Congestion costs affect all workers during shocks, while learning benefits are only for clustered workers and decay rapidly with clustering levels.

Finally, I use the estimated model to conduct three counterfactuals. First, I simulate the social optimum by imposing Pigouvian taxes that charges workers for the social externality of their entry decision. The maximum equilibrium clustering level falls. The optimal allocation features many moderately-sized clusters, which are large enough to generate learning spillovers but small enough to avoid severe congestion during hometown shocks. Second, I explore the provision of insurance against hometown shocks. By reducing workers' need to increase labor supply during adverse shocks, insurance mitigates the congestion channel. Equilibrium clustering levels nearly double, and average productivity increases by 30% relative to the baseline. Third, I show that if the platform provides better learning tools that reduce the value of location-specific knowledge, equilibrium clustering levels fall significantly.

This paper contributes to several sets of literature. First, it joins a large body of work on the value of migrant networks. Existing studies extensively document how migrant networks improve labor market outcomes by increasing employment rates, earnings, and access to information (e.g., [Borjas, 1995](#); [Munshi, 2003](#); [Antoninis, 2006](#); [Woodruff and Zenteno, 2007](#); [Bayer et al., 2008](#); [McKenzie and Rapoport, 2010](#); [Patel and Vella, 2013](#); [Munshi and Rosenzweig, 2016](#); [Giulietti et al., 2018](#); [Egger et al., 2021](#)). By contrast, much fewer studies examine the cost of migrant networks. A

key paper is [Beaman \(2012\)](#), which shows that large co-ethnic migrant communities can increase competition in job search and lower employment rates and average wages.

This paper contributes in three ways. First, using granular GPS data, I provide direct evidence of peer learning within hometown networks, offering a key microfoundation for understanding the productivity benefits of clustering. Second, I identify a new congestion cost of clustering: larger hometown networks increase income volatility by intensifying competition during correlated hometown shocks, reducing wages precisely when workers most need higher earnings. This second-moment effect differs fundamentally from existing evidence on how large migrant networks affect average employment rate and average wages, as in [Beaman \(2012\)](#).¹⁰ Third, I develop an equilibrium model to quantify the trade-offs between the learning benefits and congestion costs, and externalities on non-clustered workers.

Second, this paper contributes to the literature on migration as insurance against rural shocks. A rich set of studies shows that migrants adjust labor supply and remittances in response to adverse hometown shocks (e.g., [Lucas, 2004](#); [Yang, 2006](#); [Yang, 2011](#); [McKenzie et al., 2014](#); [Gröger and Zylberberg, 2016](#); [Joseph et al., 2018](#)). Other studies document general equilibrium effects of agricultural shocks on rural labor markets (e.g. [Jayachandran, 2006](#); [Akram et al., 2017](#)) and the role of migrants as rural insurance (e.g., [Munshi and Rosenzweig, 2016](#); [Morten, 2019](#)). A closely related paper is [Michuda \(2021\)](#), which shows that Uber drivers in Uganda increase their hours in response to adverse agricultural shocks.

I make two related contributions. First, I add consistent evidence that migrant workers adjust their labor supply in response to hometown shocks in the gig economy. The flexibility of gig work enables rapid and intensive responses.¹¹ More importantly, I identify a new general equilibrium effect that limits migration's insurance value: clustered migrants compete intensely with one another during correlated hometown shocks. While [Jayachandran \(2006\)](#) shows that rural shocks increase competition in origin labor markets and suggests migration as a solution, I demonstrate that clustering shifts this competition to destination labor markets, undermining the insurance benefits of migration.

Third, examining how clustering facilitates peer learning aligns with a comprehensive urban literature on the determinants and impacts of urban agglomeration.¹² Peer learning is a key mechanism to explain urban agglomeration (e.g. [Marshall, 1890](#); [Jaffe et al., 1993](#); [Glaeser, 1999](#); [Graham and Marvin, 2002](#); [Peri, 2002](#); [Roca and Puga, 2017](#); [Davis and Dingel, 2019](#); [Kerr and Robert-Nicoud, 2020](#); [Atkin et al., 2022](#)). I contribute by providing direct micro-level evidence

¹⁰This congestion cost can be particularly salient in gig economies, where flexible labor supply enables rapid adjustment to shocks but also generates intense competition during adverse events.

¹¹Given the rapid growth of gig economy, an expanding body of literature is investigating value of its flexibility (e.g., [Mas and Pallais, 2017](#); [Chen et al., 2019](#); [Angrist et al., 2021](#); [Michuda, 2021](#)).

¹²Urban economics literature has documented various factors leading to the agglomeration of workers, firms, and industries in urban contexts (e.g., [Rosenthal and Strange, 2001](#); [Duranton and Puga, 2004](#); [Giuliano et al., 2007](#); [Glaeser, 2010](#); [Glaeser et al., 2010](#); [Kukalis, 2010](#); [Krugman, 2011](#); [Moretti, 2012](#); [Giulietti et al., 2018](#); [Duranton and Puga, 2020](#); [Fajgelbaum and Gaubert, 2020](#)).

of peer learning and knowledge spillovers mechanisms in urban settings.¹³ Using GPS data, I show that workers transmit location-specific knowledge to one another. This spatial localization of knowledge also helps explain why agglomeration benefits may be concentrated within specific areas in some contexts.

The paper proceeds as follows. Section II describes the context. Section III shows how clustering enables peer learning and increases worker productivity. Section IV identifies the congestion cost of clustering. In section V, I build a location choice model. In Sections VI and VII, I estimate the model, measure worker utility with clustering, and discuss counterfactuals.

2 Context

2.1 Migrant Workers in the Gig Economy and Data

This paper studies the impacts of migrant networks using data on millions of rural–urban domestic migrant workers from a major food delivery platform in China. The platform operates similarly to services like *Uber Eats* and *DoorDash*. Delivery workers drive motorcycles to deliver meals, which is very common in many developing countries. Workers earn a per-delivery commission ranging from \$0.50 to \$1.50. Monthly earnings typically range from \$300 to \$1,000, an approximate linear function of the number of deliveries completed.¹⁴

I use four main datasets from the platform: (1) high-frequency GPS data tracking workers' movements every 20 seconds; (2) delivery-level record, including restaurant and consumer location and delivery fee; (3) network data linking each new worker to his referrer, who is also an existing delivery worker.¹⁵ Since referrers are likely new workers' closest peers on the platform, I will provide direct evidence of knowledge transmission within each pair of new worker and referrer; and (4) worker characteristics, including workers' county of birth. In this paper, I focus on delivery workers in five large cities in China. More details on data construction and sample selection are provided in [Online Appendix-Data Construction](#) section.

This setting offers several advantages for studying migrant networks. First, the rich GPS data allows for precise measurement of productivity and knowledge spillovers—a major empirical challenge in prior research. Second, I can capture clustering at a highly granular level by

¹³My analysis of new workers learning from past delivery experiences adds to studies estimating the impacts of on-the-job learning on worker productivity (e.g., [Thompson, 2010](#); [Levitt et al., 2013](#); [Haggag et al., 2017](#); [Mao et al., 2019](#); [Papay et al., 2020](#); [Cook et al., 2021](#)). I also highlight the additional value of referrers in the labor market: referrers transmit productive knowledge and increase new worker productivity (e.g., [Antoninis, 2006](#); [Beaman and Magruder, 2012](#); [Burks et al., 2015](#); [Dustmann et al., 2016](#); [Friebel et al., 2023](#)).

¹⁴Entry barriers of becoming a delivery workers are low. To start working, individuals download the platform's app and register by verifying their national ID. Most workers rent or purchase a motorcycle (typically around \$400), along with basic equipment such as a helmet and uniform (around \$100). Workers are assigned orders by the platform's algorithm based on real-time location rather than being tied to specific restaurants.

¹⁵Among new workers in Shanghai in 2021, 41% had referrers, and 54% of new worker-referrer pairs were from the same county of birth. When restricting the sample to workers who finished more than 50 deliveries, 60% had referrers, and 65% of those pairs came from the same counties.

measuring the number of workers coming from the same country of birth and working in the same neighborhood in each destination city. Finally, the gig economy has grown rapidly and is estimated to have 435 million workers worldwide in 2023 (Datta et al., 2023). Furthermore, the gig economy has become the top occupational choice among migrant workers due to lower entry barriers (Pew, 2021). Since this is a new economic form, where we may expect technology to have replaced many human interactions, we know much less about how workers make choices and the value of migrant networks in the gig economy.

2.2 Clustering of Migrant Workers

I begin by documenting three key facts about the extent of migrant clustering in this setting. First, 98% of delivery workers in large cities such as Beijing and Shanghai are domestic migrants, primarily born in lower-GDP provinces in western and central China.¹⁶ Second, the food delivery platform divides each large city into approximately 100–200 districts. Each district covers an area of roughly 5, km $\times$ 5, km and typically hosts 100–200 workers. GPS data show that most workers operate within a single district, completing about 70% of their daily deliveries there.¹⁷

Third, I find a high degree of clustering in workers’ location choices. I measure clustering as the share of workers from the same county of birth who work in the same district.¹⁸ The observed clustering level is 30%—substantially higher than the level expected under random assignment, which would be below 1%. Figure 1 plots a histogram of clustering levels. As a result, the largest hometown cluster accounts for around 15% of a district’s delivery workforce (Figure 2).¹⁹

Separate Work and Living Locations

The GPS data also allow me to distinguish between workers’ residential and work locations. Using delivery workers in Shanghai as an example (Figure 5 in Online Appendix), I find that most migrants live on the outskirts of the city or cluster in lower-rent housing compounds.²⁰ Same-origin migrants often share housing to reduce rent costs. However, living together does not necessarily imply that they work in the same districts.²¹

¹⁶Figure 2 and Table 2 in Online Appendix describes the pattern of migration across cities. Table 1 in Online Appendix provides demographic details.

¹⁷Districts are typically defined by drawing circles around commercial hubs (e.g., shopping malls) to minimize cross-district trips. Data also show that 85% of deliveries are within a 3 km radius.

¹⁸Hometown is defined at the county level. China has 2,844 counties. Workers’ work districts are defined using GPS data. Figure 3 in Online Appendix illustrates examples of clusters formed by workers from different hometowns.

¹⁹I also calculate clustering using the share of deliveries completed by workers from each hometown across districts (Figure 4 in Online Appendix). The delivery-based clustering levels are highly consistent with the worker-share-based measure.

²⁰Residential locations are inferred from GPS coordinates recorded before 7 a.m. or after 11 p.m. Work locations are based on the average latitude and longitude of daily deliveries.

²¹These migrant workers rarely own property in their destination cities. According to a platform-wide survey conducted in 2021, nearly 0% owned an apartment in their work city, and fewer than 3% planned to purchase one. As such, residential location decisions are not driven by housing ownership.

By separating workers' choices of work locations from residential locations, I avoid making strong assumptions to shut down other reasons for clustering. Instead, migrant workers can still enjoy various benefits of clustering by living together, such as sharing rent and food. However, the two forces examined in this paper—the learning benefits and congestion costs of clustering—directly affect labor income and are salient in affecting work location choices.

3 The Impact of Clustering on Learning and Productivity

In this section, I analyze how clustering enables knowledge spillovers through the migrant networks, leading to higher productivity and earnings. I conduct two main analyses: first, I open the black box to show what knowledge that workers share with each other; second, I use quasi-experiments based on entry bonuses to quantify the overall productivity return of clustering.

3.1 Descriptive Patterns

In the food delivery sector, faster delivery speeds allow workers to complete more orders per hour, directly translating into higher real wages. Therefore, delivery speed is a natural proxy for worker productivity.²² I begin by showing that new workers' productivity improves over time. Figure 3(a) plots new workers' average delivery speed since joining the platform. The concave shape indicates that productivity increases most rapidly in the early months.

Importantly, workers who cluster with same-origin peers seem to have steeper productivity curves. In Figure 3(b), I compare two groups of new workers: those working in districts with same-county peers (red line) and those without (blue line). New workers with hometown peers increase their delivery speed more rapidly over time. However, simple comparisons between clustered and non-clustered workers may be biased if these groups differ systematically in unobserved ways. In the next two sections, I address this concern by leveraging quasi-experimental variation to quantify the magnitude of knowledge spillovers and explore the underlying mechanisms.

3.2 Direct Evidence of Learning

3.2.1 What do workers learn from peers?

In the first analysis, I open the box of learning and identify the type of knowledge workers learn from their own or peers' experiences. Specifically, I ask whether workers can deliver faster when (1) workers themselves have previously visited the location, which captures learning by doing, and (2) their close peers have previously visited the location, which captures peer learning. As mentioned above, I use the network data and define each new worker's referrer as close peers from whom they can learn.

²²Delivery speed is calculated as delivery distance (in meters) divided by delivery duration (in minutes).

Furthermore, using granular GPS data, I decompose each delivery process into three parts: (1) time spent searching for restaurants, (2) driving speed on the main road, and (3) time spent searching for consumers.²³ This decomposition enables me to identify precisely where peer learning occurs across different stages of the delivery process.

I regress workers' productivity on indicators of whether workers themselves or their referrers have ever been assigned to visit a location before. The identification builds on the platform's order allocation algorithm, which exogenously assigns workers to visit different locations and results in workers possessing varying prior knowledge across locations and time. The corresponding regression function is as follows.

$$Y_{itab} = \alpha_1 \text{OwnVisit}_{ita} + \alpha_2 \text{RefVisit}_{ita} + \beta_1 \text{OwnVisit}_{itb} + \beta_2 \text{RefVisit}_{itb} + \delta_a + \delta_b + \gamma_{it} + \text{AveVisit}_{ita}^{\text{Own}} + \text{AveVisit}_{itb}^{\text{Own}} + \text{AveVisit}_{ita}^{\text{Ref}} + \text{AveVisit}_{itb}^{\text{Ref}} + \epsilon_{itab} \quad (1)$$

The regression is at the delivery level. For each delivery from a restaurant a to a consumer location b , I construct three dependent variables on worker productivity as outlined above. The four independent variables are: (1) whether a worker i has ever been assigned to visit the same restaurant in the past month before time t , OwnVisit_{ita} , (2) whether worker i has ever been assigned to visit the same consumer building, OwnVisit_{itb} , (3) whether i 's referrer has ever been assigned to visit either the restaurant, RefVisit_{ita} , and (4) the consumer building, and RefVisit_{itb} . I also include the location fixed effect, δ_a and δ_b , to control for inherent location characteristics. The worker $\times$ date $\times$ hour fixed effect, γ_{it} , takes care of worker and time heterogeneity, and enables me to compare the productivity across deliveries that a worker completes in the same hour but with differing prior knowledge.

A key concern is that order allocation may not be random across workers. Discussions with the algorithm team revealed that the allocation algorithm's primary input is workers' real-time GPS location, so that the platform can assign orders to workers nearby. The endogeneity concern is therefore that workers may strategically locate somewhere to receive certain types of orders. To address this, I follow the intuition from [Borusyak and Hull \(2023\)](#) and construct an average order allocation probability for each worker and referrer. Specifically, I randomly shuffle order assignments within each hour, each $2\text{km} \times 2\text{km}$ grid, and each tenure group. Repeating this exercise 100 times, I calculate the average likelihood that a worker and their referrer are assigned to visit each location under these counterfactual allocations.²⁴

By controlling for these allocation probabilities, the actual identifying assumption is that workers at the same location at the same time are close substitutes in the algorithm's assignment process. Consequently, which worker receives which order is approximately random among

²³I calculate search times by counting GPS coordinates within 150-meter radii of restaurants and consumers. Road travel time is the rest of the delivery time divided by distances.

²⁴As a robustness check, I try another method to calculate the probability of visiting a location given workers' characteristics. Specifically, I run a logit regression of visiting a location on a worker's tenure, age, gender, historical performances, whether they have a referrer, and the referrers' characteristics. The results are similar.

eligible workers at that location. Finally, workers can reject order assignments, with an average rejection rate of 9%. To address potential strategic rejection, I construct the key independent variable using initial assignments rather than actual deliveries: whether a worker has ever been assigned to visit a location, regardless of whether they accepted.²⁵

Table 1 reports the regression findings. Both learning by doing and learning from peers play important roles. A worker’s own prior visit to a location reduces search time by approximately 20% when they visit the same restaurant or consumer building again. A referrer’s prior visit decreases a new worker’s search time by around 10%, indicating knowledge transmission within this pair of workers. Conversely, travel speed on the road does not display the same patterns. These results are consistent with field insights: GPS tools help workers navigate main roads, but learning is crucial when workers navigate in narrow neighborhoods and search for specific addresses.

3.2.2 Where does learning happen?

The coefficients from this delivery-level regression reveal valuable information: the amount of learning needed in each location. I thus divide Shanghai into approximately 200 grids that are a $2km \times 2km$ each. I run the delivery-level regressions for each grid separately and compute a grid learning index based on the coefficients in front of the $OwnVisit_{ita}$ and $RefVisit_{ita}$ variables.

In Figure 4 and Table 2, I explore the correlation between these learning indexes and grid-level characteristics, including average building age and density. I find that the estimated learning magnitudes are most pronounced in old, dense, and crowded neighborhoods. This highlights the importance of learning in areas where exact locations are difficult to find.

3.3 IV Estimates of Clustering on Productivity

The preceding analysis provides direct evidence of what knowledge workers learn from peers. I next quantify the overall effect of clustering on new workers’ productivity. Estimating this relationship faces a key empirical challenge: workers’ location choices are not random and may correlate with unobserved productivity determinants. For example, workers who learn quickly through their own experience may choose not to cluster with peers. To address this concern, I construct an instrumental variable exploiting quasi-random variation from entry bonuses, which exogenously shift where new workers choose to work without directly affecting their productivity.

²⁵I also conduct balance checks, where I compare the characteristics of deliveries: (1) allocated to workers who visit a location for the first time versus those on repeat visits, (2) allocated to workers with varying tenures, and (3) accepted versus rejected by workers. I find no significant differences for these comparisons in Figure 9 and Figure 10 in [Online Appendix](#).

3.3.1 IV: bonus-predicted clustering level

Consider a new worker i , who comes from hometown h , joins the platform in week T^e . The endogenous variable is the share of hometown workers in the worker's actual work district, d .

$$\text{Share}_{hdt} = \frac{N_{hdt}}{N_{dt}}$$

where N_{hdt} is the number of workers from hometown h in district d in week t , and N_{dt} is the total number of delivery workers in district d in week t .

While new workers can choose any district in which to work, I exploit quasi-random variation in their location choices induced by entry bonuses. These entry bonuses are fixed monetary rewards offered to new workers who complete a minimum number of deliveries in bonus-target districts.²⁶ These bonuses thus act as natural experiments that exogenously incentivize new workers to enter bonus-offering districts. I also restrict to bonuses offered in the week after a worker registers, ensuring bonuses affect only district choice instead of entry decision.²⁷

To formalize the IV, as mentioned above, 60% of new workers have a referrer, an existing worker j_i who works in district d_{j_i} . I find that new workers tend to locate close to their referrers. Therefore, I define districts adjacent to one's referrer's district as a new worker's choice set, D_i^P . Data shows that 90% of new workers choose to work in one of the districts within this choice set. Given the choice set D_i^P , I know which district offers entry bonuses in week $T^e + 1$. I predict new workers will be exogenously incentivized to join the bonus-target district. In summary, the instrument variable for Share_{hdt} is the share of hometown workers in a district that offers entry bonuses in week $T^e + 1$ within each new worker's choice set, denoted as $\text{PredictShare}_{ihd_i^P t}$.

$$\text{PredictShare}_{ihd_i^P t} = \max_{d \in D_i^P} \left\{ \frac{N_{hdT^e}}{N_{dT^e}} \times \mathbb{1}\{\text{bonus}_{d,T^e+1}\} \right\}$$

The main variation of IV comes from the spatial and temporal distribution of entry bonuses across districts and weeks. While bonus offerings are endogenous to district-level labor shocks, for example, it is usually districts experiencing worker departures that offer these entry bonuses, this identification strategy does not require bonuses to be randomly assigned. I include district $\times$ week fixed effects in all specifications. This means I compare workers who work in the same district and face the same district shock, but differ in the share of hometown peers that they have in that district. Put differently, consider two new workers who join the platform in the same week and face the same entry bonuses. They choose to work in the same bonus-offering district and

²⁶The platform occasionally offers entry bonuses in specific districts and weeks to balance local labor supply and demand. In Shanghai in 2021, 72% of entry bonuses were active for only one week, and 87% of districts offered at least one bonus week during the year. In most cases, new workers are required to operate within the targeted districts for approximately a month to complete around 200 deliveries so that they can receive 400 RMB. I provide more details about the bonus in Figure 12 in [Online Appendix](#).

²⁷I plot the timeline of entry decision and bonus in Figure 13 in [Online Appendix](#)

are subject to the same district-level shocks. The only difference is that one worker luckily has hometown peers in the bonus district, while the other does not. Therefore, the identifying variation comes from differences in hometown peer distribution across bonus-offering districts within the same week, not from the presence of bonuses per se. This approach thus does not require bonuses themselves to be randomly assigned.

3.3.2 Results: Average Effect of Clustering on Productivity

I run the IV regression at the worker-week level as below.

$$Y_{iht} = \alpha \text{Share}_{hdt} + \gamma_{dt} + \gamma_{ht} + \gamma_{Te} + \text{AveShare}_{iht} + \epsilon_{iht} \quad (2)$$

As explained above, the independent variable is the share of hometown workers in each new worker's actual work district, Share_{hdt} , and the instrumental variable is the bonus-predicted share, $\text{PredictShare}_{iht}$. I also include (1) the district-week fixed effect to control for any district-level labor market fluctuations, including the heterogeneity associated with the offering of entry bonuses, (2) the hometown-week fixed effect to adjust for any hometown shocks that might influence workers' labor supply, and (3) entry cohort fixed effects to only compare the productivity of workers who enter around the same time.

A final identification concern is that both the endogenous variable (Share_{hdt}) and the instrument ($\text{PredictShare}_{iht}$) depend on existing hometown network size, N_{hdt} . Workers with larger hometown networks are mechanically more likely to have higher clustering levels regardless of which district offers entry bonuses. To address this, I follow [Borusyak and Hull \(2023\)](#) and construct an average predicted clustering level based on counterfactual bonus allocations. Specifically, I shuffle entry bonus assignments within a three-month window around each worker's entry date. Each counterfactual allocation generates a predicted clustering level. I then calculate the average predicted clustering level (AveShare_{iht}) across all bonus allocation counterfactuals. Controlling for this average predicted clustering level ensures I compare workers with similar hometown network sizes in the destination city. The actual identification assumption is that conditional on the set of districts offering bonuses in reality, which specific district offers the bonus in a given week is approximately random.

The IV results are reported in Table 3. Column 5 reports the first-stage results, where the predicted clustering level strongly correlates with the actual clustering level. Columns 6-9 present the IV results. They indicate that new workers operating in districts with a higher share of hometown peers tend to deliver food more rapidly, complete more deliveries per hour, and have higher weekly earnings. Conversely, the clustering level does not significantly affect new workers' hours. In terms of magnitude, compared to the average delivery speed, the presence of one additional same-origin peer in the same district corresponds to a 0.4% increase in new workers' productivity. Working in a district with an average clustering level results in a 5% productivity

increase for new workers.

I also conduct several robustness checks. First, table 4 in [Online Appendix](#) reports additional measures of workers' labor market performance. When workers cluster in districts with more hometown peers, they are less likely to be late for deliveries and slightly less likely to relocate to other districts. Second, while I use the number of workers to measure clustering in the baseline specification, the results in Section 3.2 suggest that the total number of deliveries completed by peers also matters. Therefore, in Table 5 in [Online Appendix](#), I construct two alternative clustering measurements: clustering weighted by peers' tenure in column (1) and clustering weighted by total deliveries completed by hometown peers in the past week in column (2). These two measurements further capture the intensive margin of peer knowledge, and they report patterns similar to the baseline specification.

3.3.3 Diminishing Marginal Gain from Clustering

More interestingly, I am able to recover the full shape of productivity improvements across clustering levels by running the IV regression with a set of indicators, $\mathbb{1}\{\text{Share}_{iht} = c\}_{c=1,\dots,20}$ and corresponding instrumental variables, $\mathbb{1}\{\text{PredictShare}_{iht} = c\}_{c=1,\dots,20}$, each representing a distinct clustering level.

$$Y_{iht} = \sum_{c=1,\dots,20} \alpha_c \mathbb{1}\{\text{Share}_{iht} = c\} + \gamma_{dt} + \gamma_{ht} + \gamma_{Te} + \text{AveShare}_{iht}^p + \epsilon_{iht} \quad (3)$$

I visualize these coefficients $\alpha_{c=1,\dots,20}$ in Figure 5. It shows that new workers achieve higher productivity as the clustering level rises, but the marginal gain decreases. Specifically, when the clustering level reaches around 10%, adding one more same-origin peer does not lead to a further increase in productivity.

3.3.4 Additional Robustness Check: Restaurant Clusters and Worker Clusters

The IV estimates show that clustering increases new workers' productivity. We may wonder whether factors unrelated to peer learning could explain this result. For example, if Cantonese restaurants co-locate with Cantonese delivery workers, new Cantonese workers might be more productive due to shared language and cultural familiarity with restaurant owners rather than peer learning. While this alternative mechanism would not bias the estimated magnitude, it would affect the interpretation of the results.

To address this concern, I directly compare the geographic distributions of restaurants and workers from the same origins. I infer restaurant origins from their brands and cuisines (e.g., "Cantonese food" or "Sichuan food"). Figure 6 shows no significant correlation between restaurant clusters and worker clusters from the same origins. This null result provides additional support for the peer learning interpretation: productivity gains arise from knowledge spillovers among

workers rather than ethnic matching with restaurants.

In summary, this section quantifies the overall impact of clustering on worker productivity. I also highlight one crucial underlying mechanism: new workers learn from peers' past delivery experiences, saving time searching for restaurants and consumer locations. Therefore, having more peers nearby expands new workers' accessible knowledge and increases average productivity.

4 The Impact of Clustering on Income Risks

In this section, I analyze the labor market congestion costs of clustering, which increase migrant workers' income volatility.

4.1 Individual Labor Supply Responses to Shocks

Existing studies have extensively documented that migrant workers tend to work longer and function as insurance against adverse shocks in rural areas (Yang, 2011). The reason is that migrant workers care about left-behind family members and aim to send remittances when adverse shocks occur in their hometowns. In this context, delivery workers are usually the primary income source for their household. Furthermore, the flexibility of the gig economy allows delivery workers to quickly adapt their labor supply in response to hometown shocks (Michuda, 2021).

To estimate how delivery workers respond to adverse hometown shocks, I use floods that occurred between June and August 2020, the worst since 1998 (Wall Street Journal, 2020).²⁸ I identify floods based on daily precipitation data. Specifically, I create a proxy for floods by identifying instances where accumulated rainfall over two days exceeds 160mm in each county and week, doubling the threshold for heavy rain.²⁹ Of 2,423 counties with precipitation data, 8% experienced flood shocks at least once during the period. Among counties with at least one active delivery worker in the sample, 11% experienced flood shocks at least once.³⁰

I run a regression at the worker-week level to identify workers' labor supply responses to these adverse hometown shocks. The specification is as follows.

$$Y_{iht} = \beta \text{Shock}_{ht} + \gamma_i + \gamma_{pt} + \gamma_d + \epsilon_{iht} \quad (4)$$

where Shock_{ht} indicates if worker i 's hometown h experienced any adverse shock in the past month. I include workers' origin province-week fixed effect, γ_{pt} , to absorb origin region-related time fluctuations, individual fixed effect, γ_i , to control for worker heterogeneity and district fixed

²⁸According to the Ministry of Emergency Management, the floods have affected 63.46 million people and caused a direct economic loss of 178.96 billion RMB, which are 12.7% and 15.5% higher than the 2015-2019 average (Post, 2020).

²⁹In most countries, the threshold for heavy rain and above is experiencing 80 millimeters per 24 hours. I also run robustness checks using different rainfall thresholds.

³⁰I plot the geographic distribution of flood shocks in Figure 14 in Online Appendix

effect γ_d , to control for district heterogeneity.³¹

Table 4 reports the effects of hometown floods on migrant workers. When their hometown experiences floods, migrants increase their labor supply by an average of six hours per week. By working longer, workers complete 17 additional deliveries, increasing weekly income by around 150 RMB. The extensive margin also responds: the share of workers completing at least one delivery increases slightly among those from affected counties.³²

I also study other forms of hometown shocks. During the study period, various cities imposed pandemic lockdowns at different points in time. Lockdowns in workers' hometowns mean that their left-behind family members could not work outside, which might also incentivize migrant workers to work longer and send remittances to support their families. As Table 6 in [Online Appendix](#) shows, when a migrant worker's hometown was under lockdown, affected migrants also tended to work harder in the destination city.

4.2 District Responses to Aggregate Shocks

I next estimate the aggregate impacts of migrant workers' hometown shocks in the destination labor market. The intuition is that consider a district with a large share of delivery workers coming from the same county of birth, h_c . If an adverse shock occurs in hometown h_c , these migrant workers may increase working hours around the same time, resulting in the total labor supply surging in the destination market.

I first construct indicators for district shocks based on the predicted share of workers experiencing adverse hometown shocks in each district and week, as below.

$$\text{ShockShare}_{dt} = \sum_h \frac{N_{hdt^0}}{N_{dt^0}} \text{Shock}_{ht}$$

where $\frac{N_{hdt^0}}{N_{dt^0}}$ is the hometown composition in each district in May 2020. I set it as fixed when computing the shock share in order to control for the endogenous changes in hometown compositions due to hometown shocks. The variation of ShockShare_{dt} thus comes from two sources: (1) different hometown compositions across districts and (2) hometown shocks that occur during different weeks.³³ I construct the indicator for aggregate district shocks as follows.

$$\text{AggShock}_{dt} = 1\{\text{ShockShare}_{dt} > 15\%\}$$

³¹Specifically, I add origin province-week fixed effects to capture different work habits for workers from different regions across the year. For example, workers from northern China may prefer to work longer during the summer.

³²Based on conversations with workers, migrant workers can remit money home through the widely available digital transfer systems in China without traveling back in person.

³³Figure 16-(a) in [Online Appendix](#) shows the histogram of district shocks. Figure 16-(b) in [Online Appendix](#) plots the correlation between hometown clustering levels and sizes of district shocks: a higher clustering level is correlated with a larger district shock. This could be rationed by that assuming that shocks are i.i.d. across hometowns, the upper bound of Shock_{dt} depends on the highest level of clustering level in the district: $\max_t \text{Shock}_{dt} \simeq \max_h \frac{N_{hdt^0}}{N_{dt^0}}$.

where I identify an aggregate shock when the predicted share of workers facing adverse hometown shocks exceeds 15% in a district. I choose this threshold for two reasons: (1) 15% is the average clustering level across districts. As discussed above, with i.i.d. hometown shocks, the shock share is highly correlated with the highest clustering level in each district; (2) the model predicts concave impacts of aggregate shocks on labor supply. Adopting a binary shock indicator facilitates interpretation and make it easier to compare with previous estimates of individual hometown shock impacts.

I run the regression at the district-week level to estimate the impacts of aggregate shocks on district-level outcomes.

$$Y_{dt} = \delta \text{AggShock}_{dt} + \gamma_d + \gamma_t + \epsilon_{dt} \quad (5)$$

where I add district fixed effect, γ_d , and week fixed effect, γ_t , to control for district heterogeneity and labor supply fluctuations across time. For the dependent variables, I calculate the total number of orders placed by consumers each week, the total number of active workers, and the total number of working hours.³⁴

Table 5 presents the results. On the labor supply side, total working hours increase substantially during aggregate district shocks, while the number of active workers remains stable. This indicates that adverse hometown shocks cause existing migrant workers to work longer rather than attracting new workers to enter in the short run. On the demand side, the total number of consumer orders does not respond significantly.³⁵ The results indicate that large aggregate shocks result in labor supply surges without corresponding increases in consumer demand. Consequently, each worker is assigned fewer deliveries per unit of time.³⁶

Furthermore, column (4) in Table 5 shows the per-delivery commission fee does not change significantly during aggregate shocks. This is because the platform does not frequently adjust commission fees for internal logistical reasons.³⁷ With constant per-delivery fees in this context, the congestion effect operates through reduced delivery assignments per worker per hour during aggregate shocks, not through lower per-delivery compensation.³⁸

4.3 Effect of Clustering on Income during Hometown Shocks

Finally, I analyze how clustering affects workers' income during adverse hometown shocks. As shown above, when a large share of workers in a district simultaneously experience adverse

³⁴Working hours add up all hours workers engage with the platform's application by opening it and taking some actions without necessarily delivering orders at that moment.

³⁵This outcome is partly due to the low delivery fee in the context. Though the increases in labor supply potentially lead to faster delivery, I do not find that consumers respond, at least in the short run.

³⁶As a robustness check, I use continuous shock shares in Table 10 in [Online Appendix](#), which displays the same pattern.

³⁷Based on conversations with the wage setting team, they only adjust per-delivery commission fee (particularly downward adjustments) at most twice a year to avoid angering workers and damaging reputation.

³⁸In other contexts where platforms can adjust wages in response to labor supply, the congestion effect may also be a result of a lower per-delivery fee.

hometown shocks, aggregate labor supply increases while consumer demand remains unchanged. This imbalance generates labor market congestion: workers compete for a fixed pool of delivery orders, resulting in fewer deliveries assigned per worker per hour. This congestion effect also imposes higher welfare costs on clustered workers, since they face individual and aggregate shocks simultaneously. This means that aggregate labor supply surges and equilibrium wages fall precisely when workers most need to work longer and earn more, amplifying their overall income volatility.

I use the following specification to capture the congestion cost of clustering.

$$Y_{ihdt} = \beta \text{Shock}_{ht} + \delta \text{AggShock}_{dt} + \alpha \text{Shock}_{ht} \times \text{AggShock}_{dt} + \gamma_i + \gamma_{pt} + \gamma_d + \epsilon_{ihdt} \quad (6)$$

where the interaction terms, $\text{Shock}_{ht} \times \text{AggShock}_{dt}$, captures the general equilibrium impacts of experiencing the individual hometown shock and aggregate district shocks at the same time. Similar to the previous two regressions, I include workers' origin province-week fixed effect, γ_{pt} , to absorb origin region-related time fluctuations, individual fixed effect, γ_i , to control for worker heterogeneity, and district fixed effect γ_d , to control for district heterogeneity.

Table 6 reports the findings. In column (3), workers increase weekly hours by around six hours on average after adverse hometown shocks, regardless of whether they experience any aggregate district shocks. The positive coefficient on Shock_{ht} in column (2) shows that affected workers complete more deliveries by working longer. However, the interaction term in column (2) reveals that when experiencing individual hometown shocks and aggregate shocks simultaneously, clustered workers complete much fewer deliveries despite working similar additional hours as non-clustered workers.

In terms of magnitudes, during adverse hometown shocks, non-clustered workers complete 20 additional deliveries by working longer, earning 200 RMB extra. On the contrary, clustered workers only complete around ten more deliveries, earning 100 RMB extra, even though they work similar hours as non-clustered workers. A back-of-the-envelope calculation shows that during adverse hometown shocks, clustered workers' real wages are 10% lower than in the normal period.

To further unpack the source of congestion, I analyze how workers' delivery patterns change during hometown-shock periods. Given the characteristics of the food delivery sector, 80% of delivery workers are active during lunch and dinner hours.³⁹ Therefore, there is limited room to further increase labor supply during these peak periods. Instead, workers facing hometown shocks extend working hours into the afternoon and late evening. With fewer consumer orders during these off-peak hours, the increased labor supply intensifies competition.⁴⁰

Finally, we may wonder whether workers can mitigate congestion costs by relocating to less competitive districts during shock period. I do not find strong evidence of such spatial adjustment.

³⁹I plot the average attendance rate per hour in Figure 17 in [Online Appendix](#).

⁴⁰I plot the changes in attendance rate when workers face negative hometown shocks in Figure 18 in [Online Appendix](#).

Column (1) of Table 8 in [Online Appendix](#) shows that neither hometown shocks nor the interaction term significantly increases workers' probability of relocating to another district. One possible explanation is that relocating to a new district decreases worker productivity, including lower delivery speed and higher rates of late deliveries in columns (2)-(4). This pattern is consistent with results in Section 3, where I show that delivery workers accumulate location-specific knowledge. Relocation thus requires costly relearning.

In summary, I analyze the congestion cost of clustering in this section. During adverse hometown shocks, migrant workers increase their weekly labor supply by six hours on average. For clustered workers, individual hometown shocks transform into aggregate shocks in the destination district, intensifying competition for consumer orders and lowering real wages in the equilibrium. As a result, clustering increases the income volatility and aggregate risk faced by migrant workers.

5 Theoretical Framework

Drawing on the empirical results from Sections 3 and 4, I develop a model to quantify the trade-off between two countervailing effects of migrant networks: the learning benefits, which raise average productivity and income, and the congestion costs, which increase income volatility. A model is essential because evaluating this trade-off requires estimating workers' risk aversion coefficient and other parameters governing their utility function. In addition, the model provides a framework for analyzing the externalities of clustering and enables counterfactual policy analysis.

5.1 Timeline

The model has four stages. In stage 1, conditional on migrating to the destination city and becoming a delivery worker, new migrant workers form expectations over the utility of working in different districts and choose to work in a district with the highest expected utility. In stage 2, negative hometown shocks are realized with some probability. In stage 3, workers choose their working hours and remittance levels to maximize utility, where they care about their own consumption, left-behind family's consumption, and their own leisure time. In Stage 4, equilibrium wages are determined through market clearing in each district.

The model assumes that workers are forward-looking and take into account both the learning benefits and congestion costs of clustering when choosing their work location. In Section 6, I provide suggestive empirical evidence for this assumption by comparing the location choices of workers from risky (high shock probability) versus safe hometowns.

5.2 Stage 1: Choices of Work Districts

Consider a new worker i from hometown h . The utility of choosing to work in district d , denoted by V_{ihd} , is given by:

$$V_{ihd} = \mathbb{E}U_{hd} + \varepsilon_{ihd} \quad (7)$$

where $\mathbb{E}U_{hd}$ is the expected utility of working in district d for any worker from hometown h , and ε_{ihd} is the idiosyncratic preference shock, following a type-1 extreme value distribution with scale, $1/\sigma$. The probability of workers from hometown h working in district d , $\pi_{hd|h}$, is as follows.

$$\pi_{hd|h} = \frac{\exp(\sigma \mathbb{E}U_{hd})}{\sum_{d'} \exp(\sigma \mathbb{E}U_{hd'})} \quad (8)$$

5.3 Stage 2 and 3: Labor Supply Decisions With and Without Hometown Shocks

After choosing a work district d , worker i decides on labor supply L_i and remittances T_i to maximize utility. I assume workers value three components: their own consumption C_i^{own} , their left-behind family's consumption C_i^{family} , and leisure time $\bar{L} - L_i$. For estimation purposes, I specify a Cobb–Douglas utility function as follows. I show in [Online Appendix-model](#) section that model predictions and propositions also hold under general utility forms.

$$\begin{aligned} \max_{L, T} \quad & (C_i^{\text{own}})^\alpha (C_i^{\text{family}})^\beta (\bar{L} - L_i)^{1-\alpha-\beta} \\ \text{s.t.} \quad & C_i^{\text{own}} + T_i = \tilde{w}_i L_i \\ & C_i^{\text{family}} = S_h + T_i \end{aligned}$$

The two budget constraints are (1) workers allocate their total earnings, $\tilde{w}_i L_i$, between their own consumption and remittances sent back to families; and (2) the left-behind family's consumption depends on the remittance received and the hometown income, S , which is assumed to be stochastic and distributed as:

$$S = \begin{cases} S^{\text{low}} & \text{with probability } P_{\text{shock}} \\ S^{\text{high}} & \text{with probability } 1 - P_{\text{shock}} \end{cases}$$

I assume that all workers from the same hometown face the same realization of S_h , while shocks across different hometowns are independent.

5.4 Stage 4: Real Wage and Market Clearing Condition

Denote that each delivery worker i completes B_i deliveries per unit of time on average. In equilibrium, the total number of deliveries completed by all workers in district d equals the

consumer-side demand for deliveries, denoted by D_d . The market clearing condition is:

$$\sum_i B_i L_i = D_d$$

Learning from Peers

As shown in Section 3, new workers can improve their delivery speed by learning from hometown peers in the same district. Let N_{hd} denote the number of workers from hometown h in district d . I assume that a worker without any hometown peers ($N_{hd} = 0$) completes B_d deliveries per unit of time, which represents a worker's baseline productivity in the district. When a worker is surrounded by N_{hd} peers, their delivery rate increases to $B_i = \delta(N_{hd})B_d$, where $\delta(\cdot)$ captures the productivity gains from clustering. Guided by the IV estimates in Section 3, I assume $\delta(N_{hd})$ is increasing and concave, with the following properties:

$$\delta(0) = 1 ; \delta'(N_{hd}) \geq 0 ; \delta''(N_{hd}) \leq 0$$

Real Wage

Substituting $B_i = \delta(N_{hd})B_d$ into the market clearing condition, I can rewrite it as:

$$\sum_h \delta(N_{hd})B_d L_{hd} N_{hd} = D_d$$

Each delivery worker earns a real wage equal to the number of deliveries completed per hour multiplied by a fixed commission fee A .⁴¹ Thus, the real wage for a worker from hometown h in district d is:

$$\tilde{w}_i = \tilde{w}_{hd} = A \times \delta(N_{hd})B_d = A \times \delta(N_{hd}) \frac{D_d}{\sum_{h'} \delta(N_{h'j}) L_{h'j} N_{h'j}}$$

5.5 Model Predictions

From the utility maximization problem and market clearing condition, I solve for a worker's optimal labor supply, real wage, and utility as a function of hometown shocks and clustering levels: $(\tilde{L}_{hd}^*, \tilde{w}_{hd}^*, u_{hd}^*) = \phi(\{S_h\}_h, \{N_{hd}\}_h)$. The model provides the following comparative statistics.

1. $\partial \tilde{L}_{hd}^* / \partial S_h < 0$: migrant workers increase labor supply when experiencing negative hometown shocks, S^{low} .
2. $\partial \tilde{w}_{hd}^* (S_h = S^{\text{high}}) / \partial N_{hd} > 0$: when there is no hometown shock, a higher clustering level leads to higher real wage due to the learning effect, as denoted by $\delta(N_{hd})$.

⁴¹The platform does not adjust this fixed fee frequently for stability reasons. I assume this commission fee is fixed during the period.

3. $\partial \tilde{w}_{hd}^* / \partial (\sum_{h'} S_{h'j} N_{h'j}) > 0$: When more workers experience negative hometown shocks simultaneously, the real wage decreases due to the congestion effect.

Proposition 1. *If $\delta(N_d) < \eta_1$ and $\delta'(N_d) < \eta_2$ ⁴², $u_{hd}^{noshock} = u_{hd}^*(S_{hd} = S^{high})$ increases with N_{hd} . $u_{hd}^{shock} = u_{hd}^*(S_{hd} = S^{low})$ first increases but then decreases with N_{hd} .*

The proof and the choice of the threshold, η_1 and η_2 , can be found in the [Online Appendix-model](#) section.⁴³

This proposition highlights the main trade-off of clustering studied in this paper. On the one hand, clustering with hometown workers enables new workers to learn from peers and achieve higher productivity: $\delta'(N_{hd}) > 0$. However, during negative hometown shocks, a high clustering level means a larger aggregate shock in the destination district. This drives down the real wage, $\tilde{w}_{hd}^*$, precisely when workers have the greatest desire to earn more, dominating the learning benefits of clustering.

Finally, I assume that workers are risk-averse, with a relative risk aversion ratio γ . The expected utility of working in district d , as denoted by $\mathbb{E}U_{hd}^{shock}(c_h)$, is as follows:

$$\mathbb{E}U_{hd} = \frac{1}{1-\gamma} \left((1 - P_{shock})(u_{hd}^{noshock})^{1-\gamma} + P_{shock}(u_{hd}^{shock})^{1-\gamma} \right) \quad (9)$$

Proposition 2. *If $\delta'(N_d) < \eta_3$, the expected utility, $\mathbb{E}U_{hd}$, first increases with N_{hd} , and then decreases. There exists an optimal clustering level, $N_{hd}^*(\gamma) = \max \mathbb{E}U_{hd}$*

The proof and the choice of the threshold, η_3 , can be found in the [Online Appendix-model](#) section.⁴⁴

This proposition highlights that for a risk-averse worker, clustering initially increases expected utility by increasing productivity. However, beyond a threshold, the congestion effect dominates: clustering increases income volatility and reduces utility. A key parameter is the worker's risk aversion coefficient γ , which governs how workers trade off higher average income against higher income volatility.

Equilibrium

I define equilibrium as a fixed point in which each worker's choice of work district is optimal, given their idiosyncratic preference shocks and the choices of all other workers. In other words, the probability distribution over district choices for new workers must be consistent with the observed

⁴² N_d is the total number of workers in the district

⁴³In a later empirical analysis, I estimate $\delta(N_d) \simeq 5\%$ and $\delta'(N_d) \simeq 0$. Along with other estimated parameters, such as estimated S^{high} and S^{low} , I find these two thresholds are satisfied.

⁴⁴In a later empirical analysis, I estimate $\delta'(N_d) \simeq 0$. Along with other estimated parameters, I find this threshold is satisfied.

distribution of existing workers across districts, which new workers use to form expectations.⁴⁵

$$\mathbb{E}\left[\frac{N_{hd}}{N_h}\right] = \frac{\exp(\sigma \mathbb{E}U(\{N_{h'd}\}_{h'}))}{\sum_{d'} \exp(\sigma \mathbb{E}U(\{N_{h'd'}\}_{h'}))} \quad (10)$$

5.6 Market Externality

Workers' choices of work location change clustering levels across districts and, in turn, affects the utility of other workers. I decompose the externalities generated by a worker's entry decision into two components:

1. Learning externality ($\Phi^{\text{learn}}(N_{hd}) = \Delta \mathbb{E}U^{\text{learn}}(N_{hd}) \times N_{hd}$): worker i can share knowledge with other workers from hometown h , leading to higher productivity.⁴⁶
2. Congestion externality ($\Phi^{\text{cost}}(N_{hd}) = \Delta \mathbb{E}U^{\text{cost}}(N_{hd}) \times N_{hd} + \Delta \mathbb{E}U^{\text{cost-other}}(N_{hd}) \times (N_d - N_{hd})$): worker i increases every worker's income variance because of the possibility of a bigger aggregate shock.⁴⁷

Proposition 3. *If $\delta'(N_{hd} = 0) < \eta_4$, the overall externality, $\Phi(N_{hd}) = \Phi^{\text{learn}}(N_{hd}) + \Phi^{\text{cost}}(N_{hd})$ is negative for any clustering level N_{hd} .*

The proof and the choice of the threshold, η_4 , can be found in the [Online Appendix-model](#) section.⁴⁸

Proposition 3 highlights that when workers choose districts to maximize their individual utility, the resulting equilibrium clustering may deviate from the social optimum. This inefficiency arises because workers do not internalize the learning and congestion externalities their decisions impose on others. I return to this point in the next section, where I quantify the magnitude of these externalities using estimated model parameters.

6 Estimation

I structurally estimate the model using empirical results in sections 3 and 4, as well as additional variation from model predictions and quasi-experiments for key parameters.

⁴⁵ $\{N_{h'd}\}_{h'}$ represents the entire hometown composition in the district d since worker i 's utility not only depends on one's hometown peers but is also affected by other hometown clusters.

⁴⁶ $\mathbb{E}U^{\text{learn}}(N_{hd})$, which is the expected utility with only the learning benefits.

⁴⁷where $\mathbb{E}U^{\text{cost}}(N_{hd})$ is the expected utility with only the congestion costs for workers from hometown h , and $\mathbb{E}U^{\text{cost}}(N_{hd})$ is the one for the rest of workers in district d .

⁴⁸In a later empirical analysis, I estimate $\delta'(N_{hd} = 0) \simeq 2$. Along with other estimated parameters, I find this threshold is satisfied.

6.1 Estimation

There are six sets of main parameters from the model: $\{\alpha + \beta, (S^{\text{high}}, S^{\text{low}}), \delta(N), P_{\text{shock}}, \gamma, \sigma\}$. I discuss the estimation approach in three steps. First, I outline the identification strategy for parameters $(\delta(N), P_{\text{shock}}, \alpha + \beta, S^{\text{high}}, S^{\text{low}})$, which quantify a worker's utility function. Second, I estimate the risk aversion parameter, γ , which governs the curvature of the expected utility function. Third, I turn to the scale parameter of the extreme value distribution, σ , which affects the dispersion in workers' location choices.

Part 1: Utility Function

$\alpha + \beta$. This Cobb–Douglas parameter reflects the elasticity of labor supply with respect to income. I estimate it using a quasi-experimental variation from "worker festivals" organized by the food delivery platform. During these events, the platform randomly selects delivery workers to receive cash prizes ranging from 50 to 1,000 RMB.⁴⁹ I estimate $\alpha + \beta$ by regressing workers' labor supply (working hours) on the prize amount received.⁵⁰ Table 7 reports the results.

$(S^{\text{high}}, S^{\text{low}})$. These two parameters represent a household's hometown income with and without adverse hometown shocks. As shown in Section 5, migrant workers increase their labor supply during negative hometown shocks (S^{low}). Given the estimated values of (α, β) , I recover S^{high} and S^{low} from workers' average labor supply during shock and non-shock periods. The intuition is that I use workers' labor supply responses to lotteries as a benchmark to infer the income equivalent of hometown shocks from observed labor supply changes.

$(\delta(N), P_{\text{shock}})$. The productivity function $\delta(N)$ is directly identified using the IV regression in Section 3 (Figure 5). I calibrate P_{shock} as the county-level flood probabilities, constructed from precipitation data between 2000 and 2020.

Part 2: Risk Aversion Coefficient (γ)

I identify γ based on the model's prediction that workers with a higher probability of adverse hometown shocks shall cluster less since they are more exposed to the congestion cost of clustering.⁵¹ Moreover, the model predicts that the more risk-averse workers there are, the larger the

⁴⁹All delivery workers can participate by submitting the phone number associated with their platform account. Typically, 500–1,000 workers win prizes in each lottery. The prize draw is random.

⁵⁰Specifically, I divide each worker's prize amount by the average hourly wage in their work district as the independent variable. The coefficient of this adjusted prize amount then maps directly to the parameter $(1 - \alpha - \beta)$.

⁵¹Table 9 in [Online Appendix](#) provides suggestive evidence consistent with the prediction. I regress the highest clustering level for each hometown on its average flood probability. It shows that workers from riskier hometowns cluster less than those from safer hometowns. This result also provides suggestive evidence that workers are aware of the labor market congestion cost of clustering when choosing work locations.

gap is. Therefore, I can use the gaps in clustering levels between workers from risky and safe hometowns to estimate the risk aversion coefficient, γ .

For implementation, I divide workers into two groups based on whether their hometowns' flood probabilities are above (risky) or below (safe) the median, and compare clustering levels across these two groups. With log-normalization and a triple-difference design, I construct a moment condition that maps directly to the concavity of the utility function.

$$\frac{\Delta \log(\pi(N_1)) - \Delta \log(\pi(N_2))}{\Delta \log(\pi(N_3)) - \Delta \log(\pi(N_2))} = \frac{\Delta u(N_1) - \Delta u(N_2)}{\Delta u(N_3) - \Delta u(N_2)} \quad (11)$$

where $\pi(N_k)$ is the average probability of new workers choosing to work in a district with a clustering level, N_k , and $\Delta u(N_k) = (u^{\text{noshock}}(N_k))^{1-\gamma} - (u^{\text{shock}}(N_k))^{1-\gamma}$ is the utility difference between periods with and without hometown shocks at clustering level N_k .

This moment condition depends only on the risk aversion coefficient γ and the utility-related parameters estimated in Part 1; the idiosyncratic preference scale parameter σ cancels out in the process. The intuition is as follows: the difference in log entry probabilities across two clustering levels, $\Delta \log(\pi(N_1)) - \Delta \log(\pi(N_2))$, depends on the joint product of the utility difference (Δu) and σ . However, by introducing a third clustering level, the relative difference across all three levels only depends on the curvature of the utility function, which is determined by γ . The full derivation is provided in [Online Appendix-model](#) section.

Part 3: Scale of Idiosyncratic Preferences (σ)

In the baseline model, workers choose work district based on the expected utility, $\mathbb{E}U(N_{hdt})$, and idiosyncratic preferences ε_{iht} . Here I extend this baseline model to include entry bonuses, B_{dt} , offered in district d and week t . To further isolate the effect of entry bonuses, I decompose expected utility into two components⁵²: $\mathbb{E}U(N_{hdt}, B_{dt}) = \mathbb{E}U(N_{hdt}) + \Delta \mathbb{E}U(B_{dt})$. This allows me to rewrite the gravity equation as:

$$\mathbb{E}\left[\frac{N_{hdt-1}^{\text{new}}}{N_{ht-1}^{\text{new}}}\right] = \frac{\exp(\sigma \mathbb{E}U(N_{hdt}) + \sigma \Delta \mathbb{E}U(B_{dt}))}{\sum_{d'} \exp(\sigma \mathbb{E}U(N_{hdt}) + \sigma \Delta \mathbb{E}U(B_{d't}))} \quad (12)$$

Since entry bonuses are exogenous, I estimate σ by running a gravity regression using $\Delta \mathbb{E}U(B_{dt})$ as the independent variable, as shown in Table 8.⁵³

Implementation. I estimate parameters, $\{S^{\text{high}}, S^{\text{low}}, \gamma\}$ together using two-step GMM. The values $\alpha + \beta$ and σ are recovered from regressions directly, as discussed above.

⁵²Simulates confirm that $\Delta \mathbb{E}U(N_{hdt}, B_{dt})$ at any level of N_{hdt} is very similar to $\Delta \mathbb{E}U(B_{dt})$ where I set $N_{hdt} = 0$.

⁵³The regression is estimated using Poisson Pseudo Maximum Likelihood (PPML), following [Sotelo, 2019](#) and [Dingel and Tintelnot, 2020](#).

6.2 Estimates and Model Fit

Table 9 presents the estimation results. All estimates fall within expected ranges. The estimated risk aversion coefficient is consistent with values commonly found in the literature, typically ranging from 1 to 10 (e.g., Swanson, 2012). The hometown income during adverse shocks is significantly lower than in normal periods, consistent with the empirical findings.

I validate the model estimates by comparing the predicted and observed distributions of entry probabilities across different clustering levels, as shown in Figure 7. Both distributions follow the same non-monotonic pattern: entry probability first increases with the clustering level and then decreases. This is consistent with the learning-congestion trade-off predicted by the model.

The exact magnitude of entry probability does not perfectly align between the two distributions. One potential reason is information friction. For example, some new workers may be unaware of where their hometown peers are located and therefore make location choices based solely on idiosyncratic preferences. To account for this, I incorporate a fraction of new workers who have no knowledge of peer locations. I proxy this fraction using the observed share of new workers without a referrer on the platform. After introducing this adjustment, the model fit improves, as shown in figure 7(c).

7 Utility, Externality, and Counterfactuals

In this section, I quantify migrant workers' utility under different clustering levels. I also discuss the market externality of clustering and simulate equilibria with and without accounting for externality. Lastly, I evaluate counterfactuals where workers receive insurance or better learning tools and analyze how workers' location choice change.

7.1 Inverted-U-shaped Utility

I first analyze how workers' expected utility changes as the clustering level increases. To separate different forces clearly, I consider a simple case: there is only one hometown cluster at level in the district, and its clustering level is N_c . Each of the remaining workers in the district comes from a different county of birth.

I plot the utility of clustered workers, $EU(N_c)$, in Figure 8. It indicates a clear inverted-U shape where the expected utility first increases with the clustering level, reaches its peak around 10%, and decreases afterward. This inverted U shape arises because learning benefits initially matter more and are dominated by congestion costs as the clustering level increases. To quantify the two effects separately, I decompose $EU(N_c) \sim EU^{\text{learn}}(N_c) + EU^{\text{cost}}(N_c)$. $EU^{\text{learn}}(N_c)$ is defined as the utility function with only the learning benefits, without congestion effect during hometown

shocks.

$$\mathbb{E}U^{\text{learn}}(N_c) = \frac{1}{1-\gamma} \left((1 - P_{\text{shock}})(u(N_c, S^{\text{high}}))^{1-\gamma} + P_{\text{shock}}(u(0, S^{\text{low}}))^{1-\gamma} \right) \quad (13)$$

$\mathbb{E}U^{\text{cost}}(N_c)$ is the utility function with only the congestion effect, without learning effect during normal periods.

$$\mathbb{E}U^{\text{cost}}(N_c) = \frac{1}{1-\gamma} \left((1 - P_{\text{shock}})(u(0, S^{\text{high}}))^{1-\gamma} + P_{\text{shock}}(u(N_c, S^{\text{low}}))^{1-\gamma} \right) \quad (14)$$

I plot both functions in Figure 8. $\mathbb{E}U^{\text{learn}}(N_c)$ is an increasing and concave curve, following the same pattern as the productivity function, $\delta(N)$, in section 3. The intuition is that the additional productivity gain from higher clustering fades away quickly after the clustering level reaches 10%. $\mathbb{E}U^{\text{cost}}(N)$ strictly decreases as the clustering level increases since a bigger cluster results in stronger congestion effects during hometown shocks.

I also quantify the utility curve for the rest of the non-clustered workers in the district.

$$\mathbb{E}U^{\text{other}}(N_c) = \frac{1}{1-\gamma} \left((1 - P_{\text{shock}})^2(u(0, S^{\text{high}}))^{1-\gamma} + (1 - P_{\text{shock}})P_{\text{shock}}(u(N_c, S^{\text{high}}))^{1-\gamma} \right) \quad (15)$$

$$+ P_{\text{shock}}(1 - P_{\text{shock}})(u(0, S^{\text{low}}))^{1-\gamma} + P_{\text{shock}}^2(u(N_c, S^{\text{low}}))^{1-\gamma} \quad (16)$$

I plot the curve in Figure 8, which also strictly decreases. These non-clustered workers do not enjoy any knowledge spillovers but only experience the congestion effect. With i.i.d. hometown shocks, non-clustered workers usually do not experience their hometown shocks and the aggregate district shocks around the same time. As a result, their utility decreases less due to the congestion costs $|\mathbb{E}U^{\text{other}}(N_c)| < |\mathbb{E}U^{\text{cost}}(N_c)|$

7.2 Externality of Clustering

Both the learning and congestion effects of clustering generate market externalities. On the one hand, knowledge spillovers have a positive externality as workers do not internalize their value of sharing knowledge with others. On the other hand, congestion effects result in workers waiting and wasting time during shock periods, producing a negative externality. Following the definition in Section 5, I quantify the positive learning and negative congestion externality across clustering levels as follows.

1. Learning externality: $\Phi^{\text{learn}}(N_c) = \Delta \mathbb{E}U^{\text{learn}}(N_c) \times N_c$
2. Congestion externality: $\Phi^{\text{cost}}(N_c) = \Delta \mathbb{E}U^{\text{cost}}(N_c) \times N_c + \Delta \mathbb{E}U^{\text{other}}(N_c) \times (N - N_c)$

I plot these curves in Figure 9. The learning externality is positive but decreases rapidly as the

clustering level increases, consistent with the concavity of the productivity function $\delta(N)$ where marginal teaching values diminish quickly. The congestion externality is negative and decreases since a larger cluster leads to more severe congestion during hometown shocks.

I also plot the total externality, $\Phi(N_c) = \Phi^{\text{learn}}(N_c) + \Phi^{\text{cost}}(N_c)$, in Figure 9. It is negative and strictly decreases, suggesting that accounting for the externalities may lead to an equilibrium with lower clustering levels than observed.⁵⁴

7.3 Counterfactual: Equilibrium with Externality

In the first counterfactual, I simulate the equilibrium to incorporate externalities into workers' district choice. I compute the total externality associated with each entry decision follow the definition of externality in section 7.3 and without allowing other workers to adjust. I solve for the equilibrium where workers receive the Pigouvian tax equal to the externality of their choices when they decide where to work. I compare it with the decentralized equilibrium, where workers choose where to work to maximize individual expected utility, $\mathbb{E}U(N_{hdt})$.

Implementation. I first simulate the number of hometowns and their size distributions based on the data. I start with a random allocation of workers across districts for each simulation. At each step, a small share of new workers arrives (as proportional to the distribution), and they choose districts following specification 7. I use an iterative procedure to solve for the fixed point, where the distributions of existing workers and new entries converge. Given the estimated parameters, the procedure is robust to varying starting conditions and identifies the same equilibrium. Notice that the equilibrium is defined as the distribution of clustering levels instead of specific hometown-district mappings, which are contingent on starting conditions. This equilibrium can also be interpreted as a stable distribution in the long run.

The first two bars in figure 10 compare the baseline decentralized equilibrium with the equilibrium with externality. Accounting for externalities significantly reduces the maximum clustering level compared to the decentralized equilibrium. Workers also create many more small clusters in the equilibrium with externalities. However, workers' average productivity does not decrease as much as the decrease of the highest clustering level. This is mainly due to the diminishing marginal productivity return of clustering. Creating small clusters instead of a large cluster allows workers to still benefit from knowledge spillovers without big congestion costs.

7.4 Counterfactual: Provide Insurance for Hometown Shocks

In the second counterfactual, I consider providing migrant workers with insurance to mitigate hometown shocks. Providing insurance eliminates correlated shocks and labor supply responses among hometown workers, which turns off the labor market congestion costs of clustering. The

⁵⁴If multiple clusters exist in one district, interactions exist across different clusters. The externality analysis follows the same logic but is more complicated to derive an analytical solution.

third bar in figure 10 reports the counterfactual. The average clustering level almost doubles relative to the decentralized equilibrium. As a result, the average productivity of new workers increases by around 30% due to the higher clustering level.

7.5 Counterfactual: Replace Network-based Learning with Technology

In the last counterfactual, I simulate equilibrium where the platform can provide workers with better technologies to substitute network-based learning. For example, the platform can collect local knowledge from experienced workers and distribute the information to all workers through a centralized system. I consider the scenario where technology can substitute 100% of peer learning. As expected, workers cluster much less, and the average productivity increases substantially since non-clustered workers can also benefit from the technology-based learning.

8 Discussion

Through several quasi-experiments and a model, I provide new empirical and theoretical evidence of two countervailing effects of migrant clustering: learning benefits and labor market congestion costs. I further examine how the two forces interact to affect workers' location choices and utility. I show that a worker's utility displays an inverted U shape with clustering levels, where utility first increases (due to learning) and then falls (due to congestion).

These results establish a crucial micro-foundation for understanding knowledge spillovers through migrant networks. Second, the congestion cost highlights a new general equilibrium cost of clustering: clustered migrants compete with each other during adverse hometown shocks, amplifying income volatility and undermining migration's insurance value. Third, by jointly quantifying both the learning benefits and the congestion costs, the analysis offers new insights into how clustering shapes migrant workers' economic outcomes and provides a framework for policies that influence migration patterns.

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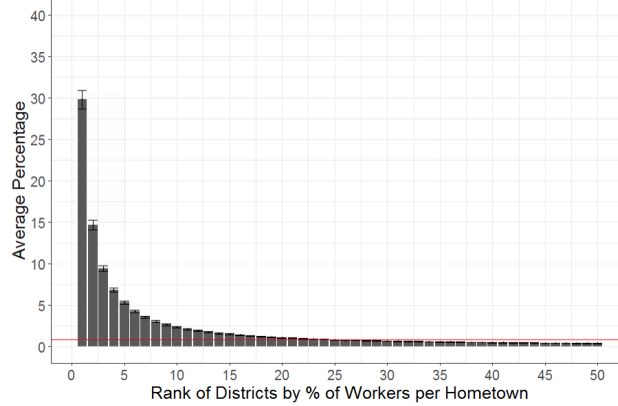
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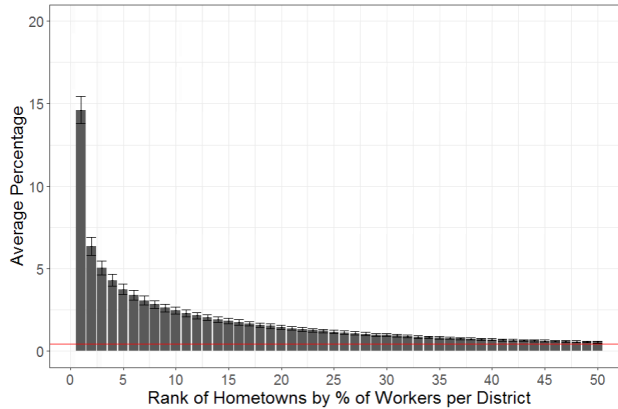
9 Figures

Figure 1: *Average Clustering Level by Sending Hometowns*



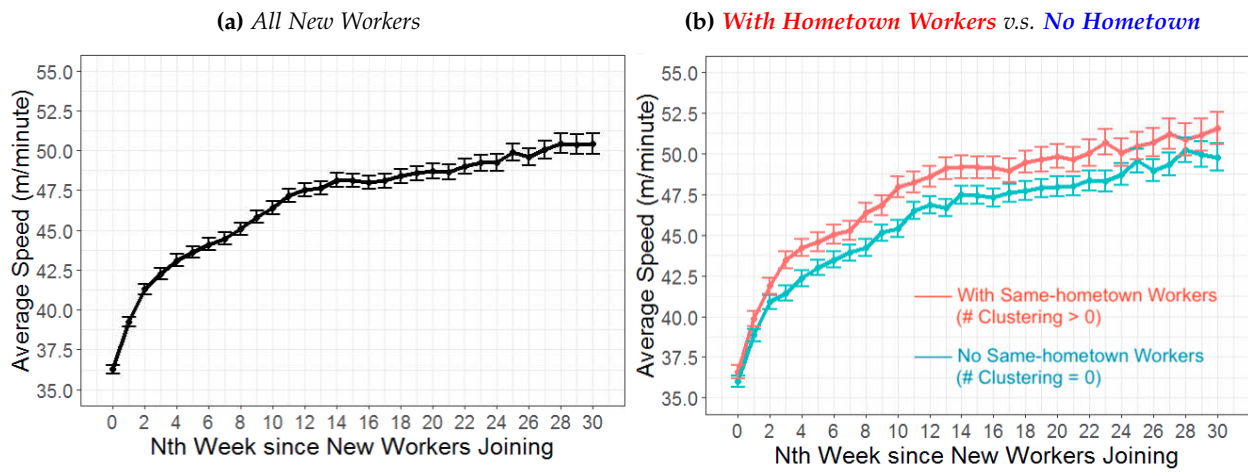
Note: This Figure shows the distribution of clustering levels by workers' hometowns. I include all active workers between 2020 and 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). For each hometown h , I calculate the share of workers in each district d : $P_{hd} = \frac{N_{hd}}{N_h} = \frac{N_{hd}}{\sum_{d \in D} N_{hd}}$. I rank districts for each hometown based on this worker share, P_{hd} . I plot the average $P_{h,d}$ by the rank of districts for each hometown. The first bar shows that 30% of workers from the same hometown cluster in one district. The red line is the simulated average share of workers in each district if workers were randomly allocated across districts.

Figure 2: *Average Clustering Level by Receiving District*



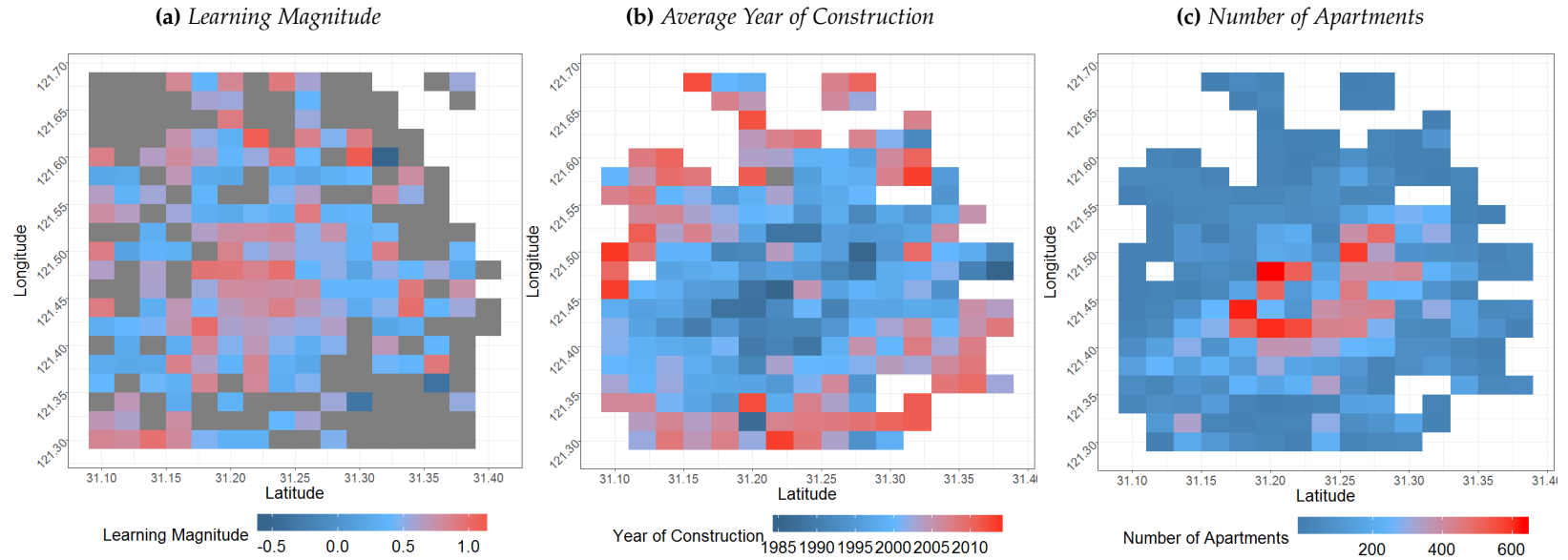
Note: This Figure plots the distribution of clustering levels by receiving districts. I include all active workers between 2020 and 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). For each district d , I calculate the share of workers from each hometown h : $S_{hd} = \frac{N_{hd}}{N_d} = \frac{N_{hd}}{\sum_{h \in H} N_{hd}}$. I rank hometowns for each district based on this share, $S_{h,d}$. I plot the average $S_{h,d}$ by the rank of hometowns for each district. The first bar shows that 15% of workers in one district come from the same hometown. The red line is the simulated average percentage of workers from each hometown if workers are randomly allocated across districts.

Figure 3: Productivity Trend of New Workers



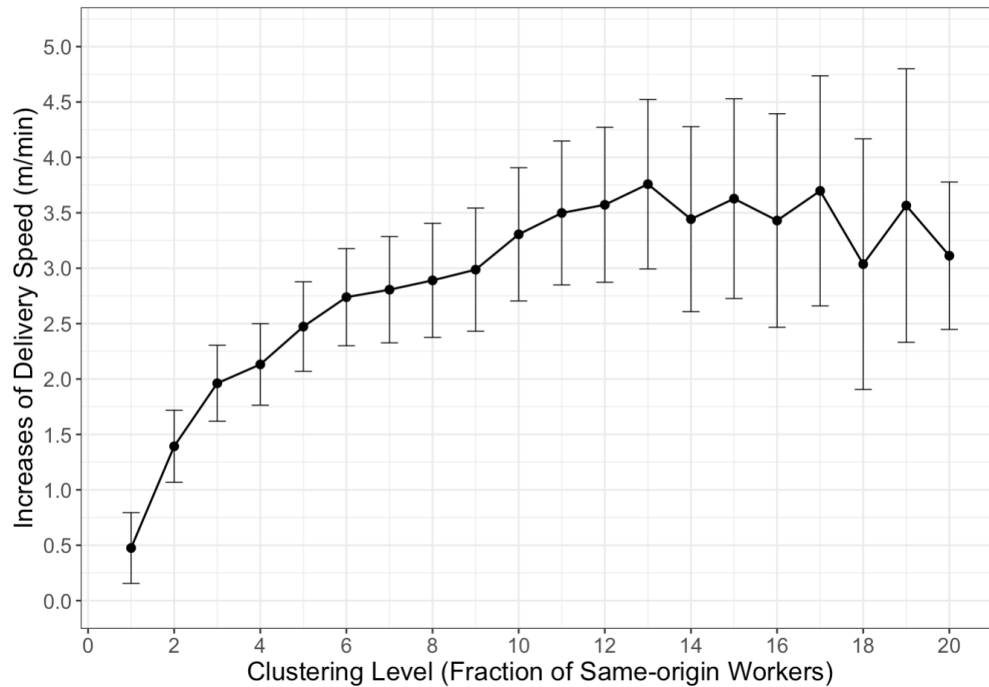
Note: This Figure plots the productivity curve of new workers. I include all new active workers in 2021 in five large cities. Details of the sample selection can be found in Appendix [Online Appendix-Data Construction Section](#). Productivity is measured by delivery speed, where I divide traveling distances (meters) by the duration (minutes). In Figure (a), I plot the productivity curve for all new workers. In Figure (b), I classify these new workers into two groups based on whether they work in a district with same-origin workers: the red line represents those with at least one same-origin worker in the same work district, and the blue line represents those without. The x-axis is the number of weeks since new workers joined the platform, and the y-axis is the average delivery speed.

Figure 4: *Heterogeneous Analysis: Learning Magnitudes across Grids in Shanghai*



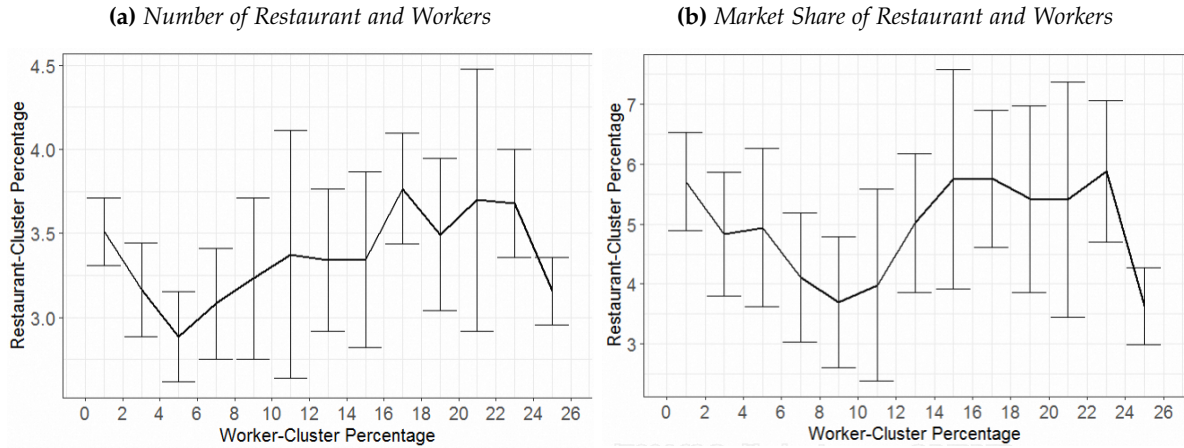
Note: This Figure plots the estimated learning magnitude in each grid and additional grid characteristics to understand where learning occurs in Shanghai. I first divide Shanghai into around 200 grids, where a grid represents a $2km \times 2km$ area. I include all new active workers between March and June 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). Figure (a) plots the coefficient from regressing new workers' search time for a location on an indicator of whether they have visited the exact location before or not, following the specification in section 3.2. I run these regressions for each grid separately and plot the coefficient in front of the indicator in Figure (a). I also obtain additional data from Lianjia.com on the average year of construction (figure (b)) and the number of available apartments (figure (c)) in each grid in Shanghai.

Figure 5: *The Productivity Gain as a Function of Clustering*



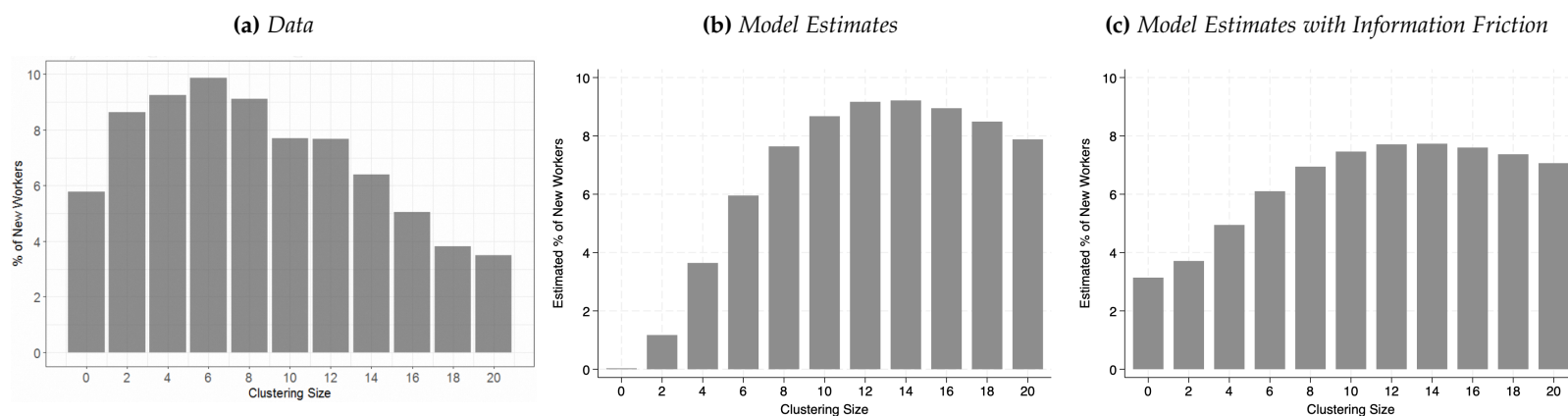
Note: This Figure plots new workers' productivity increases as a function of clustering. I include all new active workers in 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). I run an IV regression following the specification in section 3.3, where the dependent variable is new workers' average delivery speed, and the independent variables are a set of indicators for each different clustering level in the work districts. IVs are indicators for each different clustering level in the bonus-predicted districts. I plot the coefficients of these indicators in this Figure. The y-axis is the delivery speed (meter/minute). The x-axis represents clustering levels from 0% to 20%, measured by the share of hometown workers in a district.

Figure 6: *Correlation between Clustering of Restaurant and Workers from Same Origins*



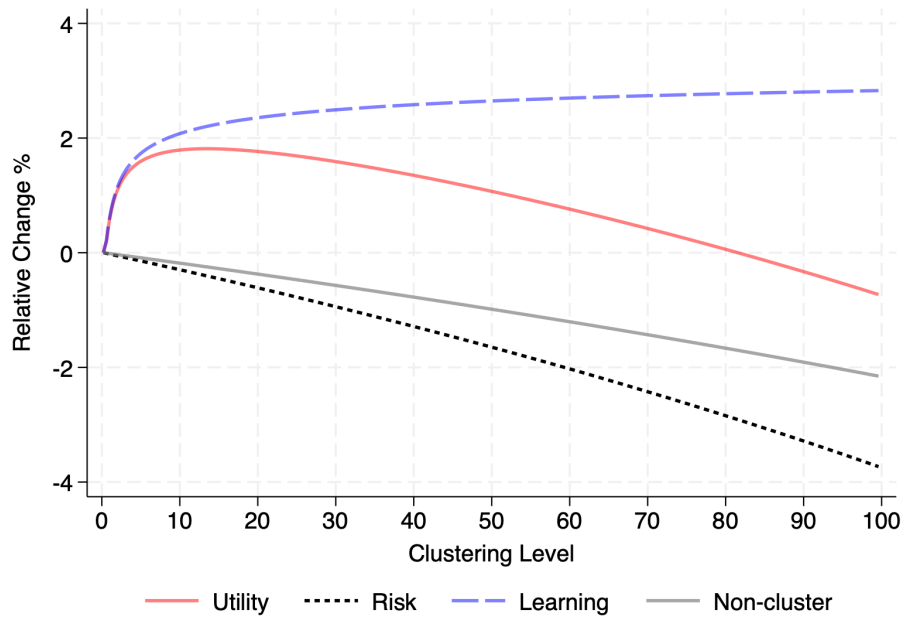
Note: This Figure plots the correlation between the share of restaurants and workers from the same origins in each district. I include all active workers between 2020 and 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). I first infer the hometown origins of restaurants based on their brands and flavors, such as Shanghai food, Sichuan food, and so on. In Figure (a), the x-axis is the share of hometown workers in each district. The y-axis is the share of restaurants from that same hometown in each district. In Figure (b), I plot the same correlation but construct the share of hometown workers and restaurants based on their sales or number of deliveries completed instead of absolute numbers in Figure (a).

Figure 7: Model Fit



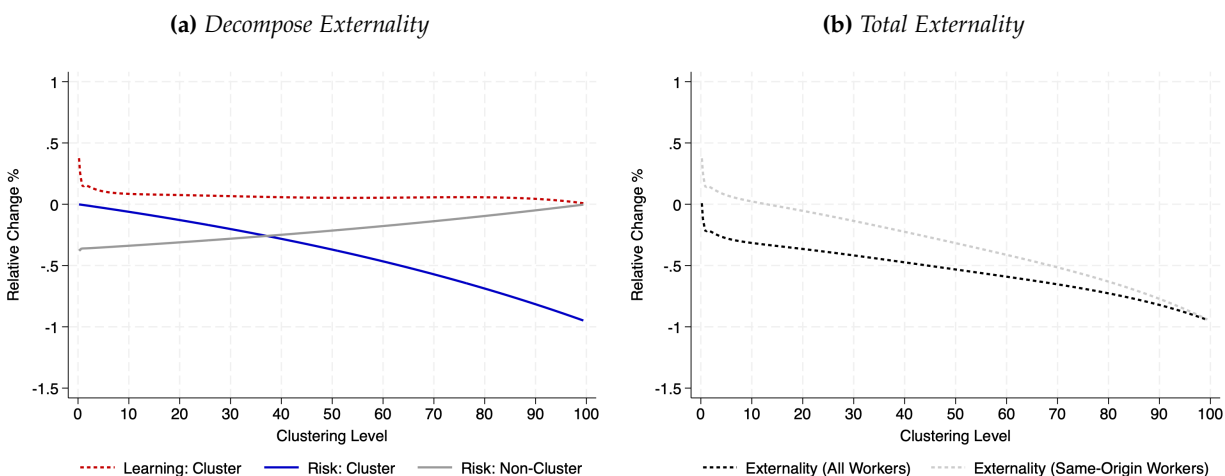
Note: This Figure plots the model fit. I plot new workers' entry probabilities across clustering levels. The x-axis is the clustering level from 0% to 20%, which is measured by the share of hometown workers in each district. The y-axis is the percentage of new workers entering districts at each clustering level. In Figure (a), I compute entry probabilities from the data, which uses the same sample as Figure 1. In Figure (b), I simulate the entry probabilities from the estimated model. Figure (c) plots the simulated probabilities after adding information frictions to the model. Specifically, I assume 40% of new workers who do not have a referrer do not have any information on existing clustering levels in the city. They thus choose districts based on their idiosyncratic preferences.

Figure 8: *Utility as a Function of Clustering*



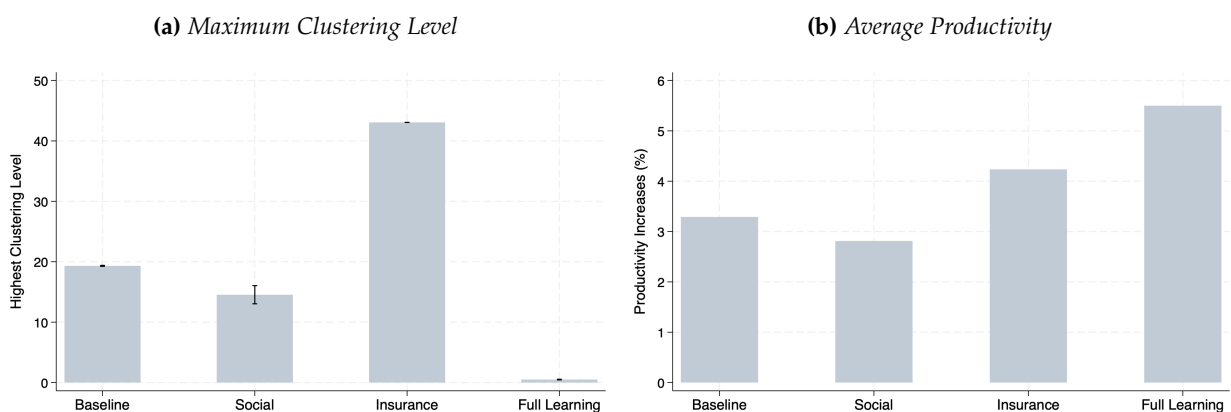
Note: This Figure plots the estimated utility with respect to clustering. The x-axis is the clustering level from 0% to 100%, representing the share of hometown workers in each district. The y-axis is the change in utility relative to the benchmark, where there is no migrant clustering. The red line represents a clustered worker's overall utility. I also decompose it into two parts: (1) the blue dashed line represents utility with only the learning benefits from clustering; (2) the black dotted line represents utility with only the congestion cost of clustering. The gray line represents the utility of non-clustered workers, who are exposed to the congestion cost of clustering.

Figure 9: Externality of Clustering



Note: This Figure plots the externality of clustering. The x-axis is the clustering level from 0% to 100%, representing the share of hometown workers in each district. The y-axis is the magnitude of externality, measured by the sum of other workers' utility changes induced by a worker's entry at each clustering level. In Figure (a), the red line represents the learning externality since a worker can share knowledge with other same-origin workers; (2) the dark blue line represents the congestion externality that a worker imposes on other same-origin workers; (3) the light blue line represents the congestion externality that a worker imposes on remaining workers from other origins. I add all three externalities together, as shown by the black line in Figure (b). The gray represents the total externality of an entry of a same-origin worker.

Figure 10: Counterfactuals



Note: I consider four scenarios: (1) "baseline" where workers choose work districts to maximize individual utility, (2) "social" where workers receive the Pigouvian tax equal to the externality of their choices when they decide where to work, (3) "insurance" where workers are provided insurance to mitigate hometown shocks, and (4) "full learning" where the platform provides better technology to substitute the entire network-based learning. Figure (a) presents the highest clustering level across the four scenarios, and figure (b) plots the average worker productivity increase.

10 Tables

Table 1: *Effects of Own and Peers' Experiences on Productivity*

	Restaurant Search Time (min) (1)	Road Driving Speed (min/km) (2)	Consumer Search Time (min) (3)
Previous Assignment - Restaurant (Own)	-1.41*** (0.04)	-0.05 (0.06)	-0.04 (0.06)
Previous Assignment - Restaurant (Referrer)	-0.47*** (0.03)	-0.04 (0.11)	-0.08 (0.09)
Previous Assignment - Consumer (Own)	-0.03 (0.04)	-0.07 (0.08)	-1.16*** (0.06)
Previous Assignment - Consumer (Referrer)	-0.06 (0.04)	-0.03 (0.10)	-0.49*** (0.06)
Date X Hour X Worker FE	Y	Y	Y
Location FE	Y	Y	Y
Borusyak & Hull Controls	Y	Y	Y
Ave. Dep. Var.	6.22	10.86	5.93
Observations	5,837,304	5,837,304	5,837,304
R ²	0.46	0.50	0.49

Notes: This Table shows regressions of productivity on four indicators: (1) whether the new worker has ever been assigned to visit a restaurant or a consumer building before; (2) whether the new worker's referrer has ever been assigned to visit the restaurant or consumer building in the last month, following the specification in section 3.2. I include all new active workers between March and June 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). Dependent variables in columns (1) and (3) are searching time (minutes) for the restaurant or the consumer building. I measure them by counting the number of GPS coordinates within a location's 150-meter radius. The dependent variable in column (2) is the average driving speed (minutes/min) between the restaurant and the consumer building. These are proxies for workers' productivity for different stages of the delivery process. I include location fixed effect and date $\times$ hour $\times$ worker fixed effect. I also control for the expected probability of a worker being assigned a delivery to visit a location by randomly re-allocating orders that are placed within the same hour, grid ($2km \times 2km$), and worker tenure group (every six months), following the intuition of [Borusyak and Hull \(2023\)](#). Standard errors are clustered two-way at the district level and date level.

Table 2: *Learning Magnitudes across Grids in Shanghai*

Dependent Variable: Learning Index α_d			
	(1)	(2)	(3)
Average Construction Year	-0.010** (0.004)		-0.008** (0.004)
Average Apartment Density		0.081*** (0.023)	0.076*** (0.027)
Ave. Dep. Var.	0.82	0.82	0.82
Observations	204	204	204
R ²	0.117	0.136	0.156

Notes: This Table shows regressions of the estimated learning magnitude on grid characteristics to understand where learning occurs in Shanghai. I include all new active workers between March and June 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). I first divide Shanghai into around 200 grids, where a grid represents a $2km \times 2km$ area. I then regress new workers' search time for a location on indicators of whether they or their referrers have visited the exact location before in each grid separately, following the specification in section 3.2. The dependent variables are coefficients from these regressions. Specifically, I calculate the learning magnitudes, $\alpha_d = (\alpha_{Ownvisit} + \alpha_{Refvisit})/2$. The independent variables are scraped from Lianjia.com, including the average year of construction across all buildings in each grid and the total number of available apartments in each grid in Shanghai. Standard errors are clustered at the district level.

Table 3: Effect of Clustering on Worker Productivity

	(1)	OLS			First Stage		IV		
		(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Ave. Speed	Deliveries Per Hour	Total Income	Working Hours	Actual Clustering Level	Ave. Speed	Deliveries Per Hour	Total Income	Working Hours
Predicted Clustering Level					0.410*** (0.004)				
Actual Clustering Level	15.109*** (0.634)	1.790*** (0.158)	358.435*** (30.481)	1.348 (1.009)		14.728*** (0.997)	1.695*** (0.224)	313.243*** (46.485)	1.250 (1.413)
District X Week FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hometown X Week FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Entry Cohort FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
BH Control	Y	Y	Y	Y	Y	Y	Y	Y	Y
Ave. Dep. Var.	46.18	4.58	682.56	24.18	2.18	46.18	4.58	682.56	24.18
Observations	417,684	417,684	417,684	417,684	417,684	417,684	417,684	417,684	417,684
R ²	0.391	0.345	0.374	0.346	0.544				
First-Stage F					9,514				

Notes: This Table shows regressions of new workers' productivity on clustering levels. I include all new active workers in 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). I report the OLS results in columns (1)-(4), where the independent variable is the share of hometown workers in each new worker's actual work district. I report the first-stage in column (5) and IV results in (6)-(8). The instrumental variable is the predicted clustering level induced by entry bonuses. Dependent variables are (1) average speed, as measured by delivery distance divided by delivery duration, (2) average number of deliveries completed per hour, (3) total weekly earnings, and (4) total weekly working hours. All dependent variables are calculated at the worker-week level. Standard errors are clustered two-way at the district level and the week level.

Table 4: *Delivery Workers' Labor Supply Responses to Hometown Shocks*

Dep. Var.: Average Weekly Outcomes			
	$\mathbb{1}\{\text{Active}\}$	Ave. Deliveries	Ave. Hours
$\mathbb{1}\{\text{Hometown Shock}\}$	0.028*** (0.005)	16.648*** (1.797)	6.095*** (0.514)
Province $\times$ Week FE	Y	Y	Y
Worker FE	Y	Y	Y
District FE	Y	Y	Y
Flood Probability Control	Y	Y	Y
Ave. Dep. Var.	0.76	78.10	22.08
Observations	638,810	638,810	638,810
R ²	0.395	0.686	0.682

Notes: This Table shows regressions of workers' weekly labor supply on hometown shocks. I include active workers between May and August 2020 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). The dependent variables measure workers' weekly performance on the platform: (1) being active (completing at least one delivery), (2) number of deliveries completed, and (3) weekly working hours. The independent variable is an indicator of whether a worker's hometown experienced a flood shock in the last four weeks. I include origin province $\times$ week fixed effects, district fixed effects, and worker fixed effects. I also control for simulated flood probability by permuting the timing of actual floods one week earlier or later for each county-week observation. Standard errors are two-way clustered at the district level and week level.

Table 5: *District-level Response to Aggregate Shocks*

Dep. Var.: Weekly Outcomes at the District Level				
	Number of Workers	Number of Orders	Total Working Hours	Per-delivery Commission Fee
1{District Shock Share > 15%}	−1.023 (1.563)	26.539 (68.666)	143.435*** (17.749)	0.022 (0.018)
Week FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Ave. Dep. Var	132.59	7,233.97	2,408.26	6.41
Observations	14,391	14,391	14,391	14,391
R ²	0.969	0.968	0.970	0.916

Notes: This Table shows regressions of district-level market performances on aggregate hometown shocks. I include active workers between May and August 2020 in five large cities. Details of the sample selection can be found in [Online Appendix](#)-Data Construction Section. Dependent variables are (1) total active workers in the district-week, (2) total deliveries completed, (3) total working hours, and (4) average per-delivery commission fee. The independent variable is an indicator for whether the predicted shock share exceeds 15%. The predicted shock share measures the expected proportion of workers experiencing hometown floods in each district-week, calculated using the hometown composition of workers in May 2020 (pre-shock baseline) rather than the contemporaneous worker composition. I include week and district fixed effects. Standard errors are clustered at the week level.

Table 6: Congestion: Impacts of Clustering on Delivery Workers' Real Wages during Shocks

Dep. Var.: Average Weekly Outcomes			
	1{ Active }	Ave. Deliveries	Ave. Hours
1{ Hometown Shock}	0.028*** (0.005)	17.775*** (1.916)	6.086*** (0.549)
1{District Shock Share > 15%}	−0.0001 (0.002)	−0.919 (0.926)	−0.272 (0.233)
Interaction Term	−0.001 (0.003)	−8.433*** (1.229)	0.110 (0.338)
Province × Week FE	Y	Y	Y
Worker FE	Y	Y	Y
District FE	Y	Y	Y
Flood Probability Control	Y	Y	Y
Ave. Dep. Var.	0.76	78.10	22.08
Observations	638,810	638,810	638,810
R ²	0.395	0.686	0.682

Notes: This Table shows regressions of workers' weekly labor supply on hometown shocks and district shocks. I include active workers between May and August 2020 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). The dependent variables measure workers' weekly performance on the platform: (1) being active (completing at least one delivery), (2) number of deliveries completed, and (3) weekly working hours. The independent variables are (1) an indicator for whether a worker's hometown experienced a flood shock in the last four weeks, (2) an indicator of district shock based on whether the predicted shock share in the district is higher than 15%, and (3) the interaction term. I include origin province × week fixed effects, district fixed effects, and worker fixed effects. I also control for simulated flood probability by permuting the timing of actual floods one week earlier or later for each county-week observation. Standard errors are two-way clustered at the district and week levels.

Table 7: Estimate $\alpha + \beta$: Labor Supply to Lottery Prizes

Dep. Var.: Weekly Working Hours		
	(1)	(2)
Lottery Amount/Average Hour Wage	-0.59*** (0.06)	-0.61*** (0.07)
Worker FE	Y	Y
Week X District FE	Y	Y
Week X Hometown FE		Y
Observations	48,126	48,126

Notes: This Table shows regressions of workers' weekly labor supply on the amount of prizes they won from the lotteries. I use lotteries offered by the platform between 2020 and 2021. I restrict the sample to delivery workers who completed at least 50 deliveries within seven days prior to the lottery and worked in a district with at least one worker winning the lottery. Dependent variables are workers' weekly working hours. The Independent variable is the prize amount divided by the average hourly wage. Standard errors are clustered two-way at the district level and week level.

Table 8: Gravity Regression

Dep. Var.: Number of Worker from Hometown h Choose District d in Week t		
	(1)	(2)
$\Delta EU(B_{dt})/\bar{EU}$ (Percent Change)	2.86*** (0.01)	3.06*** (0.02)
Week FE	Y	
Week X Hometown FE		Y
District X Hometown FE	Y	Y
Observations	309,725	303,193

Notes: This Table shows gravity regressions of new workers' district choices on entry bonuses. I include new active workers in 2021 in five large cities. Details of the sample selection can be found in [Online Appendix-Data Construction Section](#). The dependent variable is the number of new workers from each hometown joining different districts across weeks. The Independent variable is the utility change caused by the entry bonus in each district across weeks. The regression is estimated using Poisson Pseudo Maximum Likelihood (PPML), following [Sotelo \(2019\)](#) and [Dingel and Tintelnot \(2020\)](#).

Table 9: *Parameter Estimates*

	Parameter	Estimate	Conf. Interval
Labor Supply Elasticity	$\alpha + \beta$	0.39	[0.26,0.52]
High Hometown Income	S^{high}	129.79	[94.40, 152.63]
Low Hometown Income	S^{low}	-71.82	[-101.76, -33.49]
Risk Aversion Coefficient	γ	2.64	[2.03, 3.51]
Scale of Idiosyncratic Preferences	σ	3.06	[3.02, 3.10]

Notes: This Table reports the estimated parameters. Coefficients, $\alpha + \beta$ and σ are estimated by regressions as specified in section 6. I also report their estimation in table 7 and 8. Coefficients, S^{high} , S^{low} , and γ , are estimated by two-step GMM jointly. The last column reports the 95% confidence interval estimated via either regression results or a 200-round bootstrap.